# A SHARP RANK-$r$ PARTIAL-TRACE INEQUALITY VIA DOUBLE-SKEW SINGULAR-VALUE BOUNDS

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Abstract. We prove the arbitrary-matrix rank-$r$ partial-trace inequality for all ranks: for every finite-dimensional bipartite Hilbert space and every operator $C$ of rank at most $r$,

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \le r\|C\|_2^2 + \frac{1}{r}|\mathrm{Tr}\, C|^2.$$

Both coefficients are optimal — the first outright, the second once the first is fixed — whenever one factor has dimension at least $r$ and the other at least 2, and for nonzero $C$ equality forces $\operatorname{rank} C = r$. This is the extension beyond normal matrices that Costa Rico and Wolf identify as open.

The argument has two steps. A two-vector action bound on the double-skew space $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$ is proved from the Autonne–Takagi factorization, following the strategy of Fu, Gao, and Park's sharp Schmidt-rank-two projection estimate, to which that bound is equivalent. An ancilla construction and a partial transpose then reduce the rank-$r$ inequality to pair-indexed antisymmetric modes controlled by the bound: a complete-graph summation costs only the vertex degree $r-1$, and a cancellation in one distinguished direction yields the sharp trace coefficient $1/r$.

The development is machine-checked in Lean 4 in full, proved end to end. What is certified is the deduction, not the faithfulness of the formal definitions to their informal counterparts; Appendix A gives the repository, the versions, the reported axioms, and the formal–informal correspondence.

Consequences include the exact rank-constrained two-copy score curve, an exact asymmetric tensor-product score theorem with the resulting $r$-positivity classification, Kronecker-sum Ky Fan bounds, and Schmidt-number witnesses. These applications are developed in prose only; they are not part of the machine-checked development.

**AI-generation and verification disclosure.** The problem selection, prompting, and iterative direction were supplied by the author. The mathematical arguments, claimed results, applications, and exposition were generated by OpenAI's GPT-5.6 and Claude Opus 5. The Lean development of Appendix A was generated by Claude Fable 5 and Claude Opus 5 under the author's direction. It includes a proof of the Autonne–Takagi factorization, so that the development is proved end to end. What a human checked there is the interface — the definitions and the conventions they encode — while Lean checks the deductions built on it. The author did not independently derive the proofs, and no claim of independent human verification is made. The manuscript has not received independent expert review and should not be relied upon until its arguments have been checked by qualified experts.

## 1. Introduction

Costa Rico proved the rank-sensitive partial-trace inequality below for normal operators and related its unrestricted rank-two case to two-copy distillability of Werner states [2]. Costa Rico and Wolf subsequently developed partial-trace relations for general matrices, including singular-value majorization results for Kronecker sums and applications to Schmidt-number witnesses and positive maps, while identifying the extension of the same sharp rank-sensitive bound beyond normal matrices as an open problem [3]. Fu, Gao, and Park recently resolved the rank-two endpoint through a sharp estimate for the $S(2)$-norm of a double-antisymmetric projection [4]. Their estimate supplies the sharp local rank-two input, but not by itself a mechanism for combining such information across an $r$-dimensional range.

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The argument below supplies that mechanism and obtains the arbitrary-matrix inequality for every rank, unconditionally. The rank-two input is Lemma 2.1, a two-vector action bound on the double-skew space $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$, which Section 2 proves from the Autonne–Takagi factorization by the argument of [4, §2]; the reduction across an $r$-dimensional range is new. An $r$-dimensional ancilla recording an orthonormal basis of the range of $C$, a partial transpose, and a completely positive map then represent the potentially negative sector by antisymmetric modes indexed by pairs $1 \le i < j \le r$, each controlled by that bound. Grouping the modes by the vertices of the complete graph $K_r$, rather than estimating its $\binom{r}{2}$ edges independently, costs only the vertex degree $r-1$; a cancellation in the distinguished code direction then improves the coefficient of $|\operatorname{Tr} C|^2$ from 1 to the sharp $1/r$. Beyond threshold statements, the applications determine the exact rank-constrained score curves — the minimum of the normalized two-copy filter form over nonzero matrices of rank at most $r$, as in Equation (49) — for both symmetric and asymmetric two-copy filters; the asymmetric formula yields a complete $r$-positivity classification for the corresponding tensor-product maps.

The Werner-state and distillability terminology used in the applications follows the standard conventions [9, 6]. Throughout, $\|\cdot\|_2$ denotes the Hilbert–Schmidt, or Schatten-2, norm. The notation $\operatorname{Tr}_U$ means the partial trace *over* the factor $U$. Inner products are conjugate-linear in the first argument and linear in the second. Relative to chosen orthonormal bases, vectorization is normalized by

$$\operatorname{vec}(|x\rangle\langle y|) = x \otimes \overline{y}.$$

The symbol $\mathrm{SR}(z)$ denotes the Schmidt rank of a vector $z$, and $\mathrm{SN}(\sigma)$ denotes the Schmidt number of a positive operator or state $\sigma$, always across the bipartition indicated in context. With the above normalization, a vector has Schmidt rank at most $k$ across the output–input cut exactly when it is the vectorization of a matrix of rank at most $k$; that is,

$$\mathrm{SR}\big(\operatorname{vec}(A)\big) = \operatorname{rank} A, \tag{1}$$

a fact used repeatedly below. Finally, for an operator on a twofold tensor power, $\operatorname{Tr}_1$ denotes the partial trace *over* the first local factor, leaving an operator on the second, and $\operatorname{Tr}_2$ traces over the second.

The paper is self-contained modulo one classical ingredient, the Autonne–Takagi factorization of a complex symmetric matrix [5, Cor. 4.4.4], from which Lemma 2.1 is proved in Section 2 following the strategy of [4, §2]; so Lemma 2.1 is not imported from their work but proved, and the formal development of Appendix A proves the Autonne–Takagi factorization as well, making that chain complete end to end. Remark 2.2 records that Lemma 2.1 is *equivalent* to the sharp projection estimate of Fu, Gao, and Park, so their Theorem 2.4 may be substituted for the proof given here, and conversely.

**Theorem 1.1** (Rank-$r$ partial-trace inequality)**.** *Let $U$ and $V$ be finite-dimensional complex Hilbert spaces. If $C \in \mathcal{L}(U \otimes V)$ is nonzero and $r_0 := \operatorname{rank} C$, then*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le r_0 \|C\|_2^2 + \frac{1}{r_0} |\operatorname{Tr} C|^2. \tag{2}$$

*Consequently, for every $r \in \mathbb{N}_{\ge 1}$ such that $\operatorname{rank} C \le r$,*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le r \|C\|_2^2 + \frac{1}{r} |\operatorname{Tr} C|^2. \tag{3}$$

*The assertion is immediate when $C = 0$.*

*Remark* 1.2 (Optimality of the coefficients)*.* The coefficients are optimal whenever $\dim U \ge r$ and $\dim V \ge 2$, or with $U$ and $V$ interchanged. More precisely, the coefficient $r$ of $\|C\|_2^2$ is optimal, and conditional on choosing this optimal coefficient, the coefficient $1/r$ of $|\operatorname{Tr} C|^2$ is optimal.

Indeed, choose

$$C = P_r \otimes |v\rangle\langle w|,$$

where $P_r$ is a rank-$r$ projection on $U$ and $v, w \in V$ are unit vectors. Then

$$\begin{aligned}\|\mathrm{Tr}_U C\|_2^2 &= r^2,\\ \|\mathrm{Tr}_V C\|_2^2 &= r|\langle w, v\rangle|^2,\\ \|C\|_2^2 &= r,\\ |\mathrm{Tr}\, C|^2 &= r^2|\langle w, v\rangle|^2.\end{aligned}$$

Thus equality holds in Equation (3), and the displayed extremizer has rank $C = r$ exactly. More generally, for nonzero $C$ equality in the rank-$r$ inequality requires rank $C = r$; the passage from the exact-rank inequality to a strictly larger rank parameter is strict. This is why the sharpness example is built from a rank-$r$ projection. For a bound of the form

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \le a\|C\|_2^2 + b|\mathrm{Tr}\, C|^2,$$

taking $v \perp w$ forces $a \ge r$. Once $a = r$ is fixed, taking $v = w$ forces $b \ge 1/r$.

The proof of Theorem 1.1, and of every result it depends on, has been formalized in Lean 4, as have the statements of Remark 1.2 and Lemma 4.7. Appendix A records the repository, the toolchain and Mathlib pins, the reported axioms, and the correspondence between the formal and informal statements.

## 2. Double-Skew Estimates and the Marginal Operator Inequality

We first prove the double-skew action bound used below. The route is that of [4, §2], with the Autonne–Takagi factorization in place of their Theorem 2.4; Remark 2.2 records that the two are equivalent.

**Lemma 2.1** (Double-skew action bound)**.** *Fix orthonormal bases of $U$ and $V$, and let*

$$\mathfrak{so}(U) := \{L \in \mathcal{L}(U) : L^T = -L\}, \qquad \mathfrak{so}(V) := \{M \in \mathcal{L}(V) : M^T = -M\}.$$

*For every $K$ in the linear span of*

$$\{L \otimes M : L \in \mathfrak{so}(U),\ M \in \mathfrak{so}(V)\},$$

*and every orthonormal pair $x, y \in U \otimes V$,*

$$\|Kx\|^2 + \|Ky\|^2 \le \frac{1}{2}\|K\|_2^2. \tag{4}$$

*Proof.* Index operators on $U \otimes V$ by pairs relative to the fixed bases, writing $N_{(a,b),(a',b')}$, and define the two partial transposes

$$(N^{T_U})_{(a,b),(a',b')} := N_{(a',b),(a,b')}, \qquad (N^{T_V})_{(a,b),(a',b')} := N_{(a,b'),(a',b)},$$

whose composite is the ordinary transpose. Both are self-adjoint involutions for the Hilbert–Schmidt inner product and they commute, so

$$\Pi := \tfrac{1}{2}(1 - T_U)\, \tfrac{1}{2}(1 - T_V)$$

is the orthogonal projection of $\mathcal{L}(U \otimes V)$ onto $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$. Hence $\Pi K = K$; and since a tensor product of two skew-symmetric matrices is symmetric, also $K^T = K$.

*Step 1: a symmetric Schmidt decomposition.* Because $K^T = K$, the Autonne–Takagi factorization [5, Cor. 4.4.4] supplies an orthonormal basis $w_1, \dots, w_n$ of $U \otimes V$ and reals $s_i \ge 0$ with

$$K = \sum_i s_i\, w_i w_i^T. \tag{5}$$

The $w_i w_i^T$ are themselves orthonormal in the Hilbert–Schmidt inner product, because $\langle uu^T, vv^T\rangle_{\mathrm{HS}} = \langle u, v\rangle^2$.

*Step 2: the two-term bound.* Write $M_u$ for the $\dim U \times \dim V$ coefficient matrix of $u \in U \otimes V$, so that $\|M_u\|_2 = \|u\|$. Carrying out the two inner sums separately gives the flip trace identity

$$\langle uu^T, (vv^T)^{T_U} \rangle_{\mathrm{HS}} = \operatorname{Tr}\big((M_u^* M_v)^2\big). \tag{6}$$

Let $N := s_1 w_1 w_1^T + s_2 w_2 w_2^T$ with $s_1, s_2 \geq 0$, write $M_i := M_{w_i}$ and $c_i := \|M_i^* M_i\|_2$. Expanding by Equation (6), using that $M_i^* M_i$ is self-adjoint for the diagonal terms and that the two cross terms are complex conjugates,

$$\langle N, N^{T_U} \rangle_{\mathrm{HS}} = s_1^2 c_1^2 + 2 s_1 s_2 \operatorname{Re} \operatorname{Tr}\big((M_1^* M_2)^2\big) + s_2^2 c_2^2.$$

Two applications of Cauchy–Schwarz for $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$, namely

$$|\operatorname{Tr}(X^2)| = |\langle X^*, X \rangle_{\mathrm{HS}}| \leq \|X\|_2^2, \qquad \|A^* B\|_2^2 = \langle AA^*, BB^* \rangle_{\mathrm{HS}} \leq \|A^* A\|_2 \, \|B^* B\|_2,$$

the second using $\|AA^*\|_2 = \|A^* A\|_2$, bound the cross term below by $-2 s_1 s_2 c_1 c_2$. Therefore

$$\langle N, N^{T_U} \rangle_{\mathrm{HS}} \geq (s_1 c_1 - s_2 c_2)^2 \geq 0. \tag{7}$$

Since $N^T = N$ forces $N^{T_V} = N^{T_U}$, expanding $\Pi$ leaves only two distinct terms, and Equation (7) gives

$$\langle N, \Pi N \rangle_{\mathrm{HS}} = \tfrac{1}{2}\big(\|N\|_2^2 - \langle N, N^{T_U} \rangle_{\mathrm{HS}}\big) \leq \tfrac{1}{2} \|N\|_2^2. \tag{8}$$

This is the only place the hypothesis $\dim \leq 2$ of [4, Lem. 2.3] is used, and Equation (7) is that lemma.

*Step 3: truncation.* Fix $a \neq b$ and put $K_{ab} := s_a w_a w_a^T + s_b w_b w_b^T$. Orthonormality of the $w_i w_i^T$ gives $\|K_{ab}\|_2^2 = \langle K, K_{ab} \rangle_{\mathrm{HS}} = s_a^2 + s_b^2 =: A$. As $\Pi$ is a self-adjoint idempotent with $\Pi K = K$,

$$A = \langle K, K_{ab} \rangle_{\mathrm{HS}} = \langle \Pi K, K_{ab} \rangle_{\mathrm{HS}} = \langle K, \Pi K_{ab} \rangle_{\mathrm{HS}} \leq \|K\|_2 \, \|\Pi K_{ab}\|_2,$$

while $\|\Pi K_{ab}\|_2^2 = \langle K_{ab}, \Pi K_{ab} \rangle_{\mathrm{HS}} \leq \frac{1}{2} A$ by Equation (8). Squaring the first display and substituting the second yields $A^2 \leq \frac{1}{2} A \|K\|_2^2$. If $A = 0$ there is nothing to prove; otherwise one factor of $A$ cancels, and

$$s_a^2 + s_b^2 \leq \tfrac{1}{2} \|K\|_2^2 \qquad \text{for } \textit{every} \text{ pair } a \neq b. \tag{9}$$

*Step 4: from pairs to the action.* By Equation (5) and orthonormality, $K^* K = \sum_i s_i^2 |\overline{w_i}\rangle\langle\overline{w_i}|$, so for an orthonormal pair $x, y$,

$$\|Kx\|^2 + \|Ky\|^2 = \sum_i s_i^2 \theta_i, \qquad \theta_i := |\langle \overline{w_i}, x \rangle|^2 + |\langle \overline{w_i}, y \rangle|^2.$$

Each $\theta_i$ equals $\|P \overline{w_i}\|^2$ for the rank-two orthogonal projection $P$ onto $\operatorname{span}\{x, y\}$, so $\theta_i \in [0, 1]$; and $\sum_i \theta_i = \operatorname{Tr} P = 2$ because $\{\overline{w_i}\}$ is an orthonormal basis. For weights of this kind and any $c_i \geq 0$, letting $a, b$ index the two largest $c_i$ and bounding every remaining term by $c_b \theta_i$,

$$\sum_i c_i \theta_i \leq c_a \theta_a + c_b (2 - \theta_a) = 2 c_b + \theta_a (c_a - c_b) \leq c_a + c_b.$$

Applying this with $c_i = s_i^2$, and then Equation (9) to that pair, gives Equation (4). □

*Remark* 2.2 (Equivalence with the Fu–Gao–Park estimate). Lemma 2.1 is equivalent to [4, Thm. 2.4], so that estimate may be substituted for the proof above.

Label the four vectorization factors $U_{\mathrm{out}}, V_{\mathrm{out}}, U_{\mathrm{in}}, V_{\mathrm{in}}$ as $1, 2, 3, 4$; then the antisymmetrized pairs $(U_{\mathrm{out}}, U_{\mathrm{in}})$ and $(V_{\mathrm{out}}, V_{\mathrm{in}})$ are Fu–Gao–Park's $(1, 3)$ and $(2, 4)$, the cut $(U_{\mathrm{out}} V_{\mathrm{out}}) : (U_{\mathrm{in}} V_{\mathrm{in}})$ is their 12:34, and vec intertwines $\Pi$ with $Q_- := P^{(1,3)}_{\wedge^2 U} \otimes P^{(2,4)}_{\wedge^2 V}$. Their estimate states that, for every $z$ of Schmidt rank at most two across that cut,

$$\langle z, Q_- z \rangle \leq \tfrac{1}{2} \|z\|^2, \tag{10}$$

both sides being homogeneous of degree two, so that no normalization is needed. By Equation (1) this says exactly that $\langle M, \Pi M \rangle_{\mathrm{HS}} \leq \frac{1}{2} \|M\|_2^2$ for every $M \in \mathcal{L}(U \otimes V)$ with $\operatorname{rank} M \leq 2$.

*Their estimate implies Lemma 2.1.* Let $P = |x\rangle\langle x| + |y\rangle\langle y|$ and $A = \|KP\|_2^2$. Since $P^2 = P = P^*$, $\|Kx\|^2 + \|Ky\|^2 = A$ and $\operatorname{rank}(KP) \le 2$; and as $\Pi$ is a self-adjoint idempotent fixing $K$, $A = \langle K, KP\rangle_{\mathrm{HS}} = \langle K, \Pi(KP)\rangle_{\mathrm{HS}} \le \|K\|_2 \|\Pi(KP)\|_2$, while $\|\Pi(KP)\|_2^2 \le \frac{1}{2}A$. Cancelling one factor of $A$ gives Equation (4). (A block-zero extension reduces unequal local dimensions to the square case they state, since extending a skew-symmetric matrix by zero preserves skew-symmetry and the Hilbert–Schmidt norm.)

*Conversely.* Given $M$ with $\operatorname{rank} M \le 2$, choose an orthonormal pair $v_1, v_2$ spanning a subspace that contains the row space $\operatorname{ran} M^* = (\ker M)^\perp$, and set $u_i := Mv_i$; then $M = \sum_{i\le 2} |u_i\rangle\langle v_i|$ and $\|M\|_2^2 = \sum_i \|u_i\|^2$. (If $\dim(U \otimes V) = 1$ then $\Pi = 0$ and there is nothing to prove.) Put $K := \Pi M$. Then $\|K\|_2^2 = \langle K, M\rangle_{\mathrm{HS}} = \sum_i \langle Kv_i, u_i\rangle \le \big(\sum_i \|Kv_i\|^2\big)^{1/2} \|M\|_2 \le \big(\frac{1}{2}\|K\|_2^2\big)^{1/2} \|M\|_2$ by Equation (4), whence $\langle M, \Pi M\rangle_{\mathrm{HS}} = \|\Pi M\|_2^2 \le \frac{1}{2}\|M\|_2^2$.

Both directions are machine-checked; see Appendix A.

2.1. **The amplification theorem.** Lemma 2.1 is one instance of a general mechanism, and it is cleaner to state the general form first and specialize. What the argument of Section 3 consumes is a single number attached to a completely positive map.

**Definition 2.3** (Rank-two Choi constant)**.** *Let $\Phi$ be completely positive on $\mathcal{L}(H)$ with Kraus family $\{A_a\}$, and let*

$$J(\Phi) := \sum_a |\operatorname{vec} A_a\rangle\langle \operatorname{vec} A_a|$$

*be its Choi operator. Set*

$$\beta_2(\Phi) := \tfrac{1}{2}\|J(\Phi)\|_{S(2)} = \tfrac{1}{2}\sup\{\langle z, J(\Phi)z\rangle : \mathrm{SR}(z) \le 2,\ \|z\| = 1\}.$$

$J(\Phi)$ is the same operator for every Kraus family of $\Phi$, so $\beta_2$ depends on $\Phi$ alone — no minimality and no orthogonality of the family is required. It is linear in $\Phi$, so a map and its constant always carry the same scale, and no normalization convention has to be tracked. By Remark 2.2, $J(\Lambda_U \otimes \Lambda_V)$ is $4Q_-$ and $\|Q_-\|_{S(2)} = \frac{1}{2}$, so

$$\beta_2(\Lambda_U \otimes \Lambda_V) = 1. \tag{11}$$

**Theorem 2.4** (Amplification)**.** *Let $\Phi$ be completely positive on $\mathcal{L}(U \otimes V)$ with* transpose-symmetric *Kraus operators, $A_a^T = A_a$. Then for every $r \ge 1$ the map*

$$\Psi_{r,\Phi} := \Phi \circ \tau + (r-1)\beta_2(\Phi)\left(\Delta - \tfrac{1}{r}\mathrm{id}\right)$$

*is $r$-positive, where $\Delta(X) = \operatorname{Tr}(X)I$ and $\tau$ is transposition. Equivalently, in the notation of Proposition 2.5, for every orthonormal family $e_1, \dots, e_r$ in $U \otimes V$,*

$$(\Phi \otimes \mathrm{id}_Q)(P_-) \preceq (r-1)\beta_2(\Phi)\,(I_{UVQ} - \Pi_e), \tag{12}$$

$$(\Phi \otimes \mathrm{id}_Q)(\rho_0^{T_{UV}}) + (r-1)\beta_2(\Phi)\,(I_{UVQ} - \Pi_e) \succeq 0. \tag{13}$$

The proof occupies Section 3; Equation (12) is Equation (22) with $\beta_2$ in place of 1, and Equation (13) follows from it because $P_+$ survives $\Phi$ and $P_+ - P_-$ is twice the partial transpose. Two hypotheses do all the work. The transpose symmetry is what puts the frame in $\delta_e^\perp$ (Equation (31)), and Remark 2.6 identifies it as a $\mathbb{Z}/2$-degree condition; the constant $\beta_2$ enters through a single Cauchy–Schwarz over the Kraus index (Equation (28)). Nothing else about $\Phi$ is used — in particular nothing about $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$ beyond Equation (11).

At $\Phi = \Lambda_U \otimes \Lambda_V$ and $\beta_2 = 1$, Theorem 2.4 is the operator inequality we now record in the form Section 3 uses.

**Proposition 2.5** (Rank-$r$ marginal operator inequality). *Let $Q = \mathbb{C}^r$, let $e_1, \dots, e_r$ be an orthonormal family in $U \otimes V$, and let $q_1, \dots, q_r$ be an orthonormal basis of $Q$. Define the unnormalized code vector and its rank-one operator by*

$$\delta_e := \sum_{i=1}^{r} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|, \qquad \Pi_e := \frac{1}{r}\rho_0.$$

*Thus $\Pi_e$ is the orthogonal projection onto* $\operatorname{span}\{\delta_e\}$*. Write*

$$\rho^0_{UQ} := \operatorname{Tr}_V \rho_0, \qquad \rho^0_{VQ} := \operatorname{Tr}_U \rho_0.$$

*Then*

$$H_r^{(0)} := rI_{UVQ} + \Pi_e - \iota_{UQ}(\rho^0_{UQ}) - \iota_{VQ}(\rho^0_{VQ}) \succeq 0. \tag{14}$$

*For $S \subseteq \{U, V, Q\}$, the map*

$$\iota_S : \mathcal{L}\left(\bigotimes_{X \in S} X\right) \longrightarrow \mathcal{L}(U \otimes V \otimes Q)$$

*uses the order inherited from $U, V, Q$: it tensors with identities on omitted factors and conjugates by the unique permutation unitary that restores the order $U \otimes V \otimes Q$. For example, $\iota_{UQ}(X)$ is the operator that acts as $X$ on the labeled $U, Q$ factors and as the identity on $V$.*

*Proof.* Fix orthonormal bases of $U$ and $V$, relative to which transposition and the skew-symmetric operator spaces are defined. Although the proof uses these bases, the resulting operator inequality is basis independent.

Expanding $\rho_0$ and transposing the $UV$ factor gives

$$\begin{aligned}\rho_0^{T_{UV}} &= \sum_{i,j=1}^{r} \left(|e_i\rangle\langle e_j|\right)^T \otimes |q_i\rangle\langle q_j| \\ &= \sum_{i,j=1}^{r} |\overline{e}_j \otimes q_i\rangle\langle \overline{e}_i \otimes q_j|. \end{aligned} \tag{15}$$

Let

$$S := \operatorname{span}\{\overline{e}_1, \dots, \overline{e}_r\},$$

and let $J : Q \to S$ be the unitary determined by $Jq_i = \overline{e}_i$. Define a self-adjoint unitary $F_J$ on $S \otimes Q$ by

$$F_J(s \otimes q) := Jq \otimes J^{-1}s.$$

In particular,

$$F_J(\overline{e}_i \otimes q_j) = \overline{e}_j \otimes q_i.$$

Thus the sum in Equation (15) is $F_J$ on $S \otimes Q$. Let

$$P_\pm := \frac{1}{2}(I_{S\otimes Q} \pm F_J)$$

be its positive and negative eigenspace projections, extended by zero on $(S \otimes Q)^\perp$; thus $P_+ + P_-$ is the orthogonal projection onto $S \otimes Q$, not the identity on $U \otimes V \otimes Q$. Then

$$\rho_0^{T_{UV}} = P_+ - P_-. \tag{16}$$

Explicitly,

$$P_- = \sum_{1 \le i < j \le r} |\eta_{ij}\rangle\langle\eta_{ij}|,$$

where

$$\eta_{ij} = \frac{1}{\sqrt{2}}(\overline{e}_i \otimes q_j - \overline{e}_j \otimes q_i). \tag{17}$$

For a finite-dimensional Hilbert space $E$, define

$$\Lambda_E(X) := \mathrm{Tr}(X)I_E - X^T.$$

If $L_{ab} = E_{ab} - E_{ba}$, then

$$\Lambda_E(X) = \sum_{a<b} L_{ab} X L_{ab}^*. \tag{18}$$

Thus $\Lambda_E$ is completely positive. Equivalently, its Choi operator is

$$J(\Lambda_E) = \sum_{i,j} |i\rangle\langle j| \otimes \Lambda_E(E_{ij}) = I - F,$$

whose spectrum is contained in $\{0, 2\}$.

Define

$$\Phi := \Lambda_U \otimes \Lambda_V \otimes \mathrm{id}_Q.$$

For an operator $Y$ on $U \otimes V \otimes Q$, direct expansion gives

$$\begin{aligned} \Phi(Y) = \iota_Q(\mathrm{Tr}_{UV} Y) &- \iota_{VQ}\Big((\mathrm{Tr}_U Y)^{T_V}\Big) \\ &- \iota_{UQ}\Big((\mathrm{Tr}_V Y)^{T_U}\Big) + Y^{T_{UV}}. \end{aligned} \tag{19}$$

Applying Equation (19) to $Y = \rho_0^{T_{UV}}$ gives

$$\Phi(\rho_0^{T_{UV}}) = I_{UVQ} - \iota_{UQ}(\rho_{UQ}^0) - \iota_{VQ}(\rho_{VQ}^0) + \rho_0. \tag{20}$$

Using Equation (16), we therefore obtain

$$H_r^{(0)} = \Phi(P_+) + \Big[(r-1)(I_{UVQ} - \Pi_e) - \Phi(P_-)\Big]. \tag{21}$$

Since $\Phi(P_+) \succeq 0$, it remains to prove

$$\Phi(P_-) \preceq (r-1)(I_{UVQ} - \Pi_e). \tag{22}$$

Let $\{L_\alpha\}_{\alpha \in \mathcal{A}_U}$ and $\{M_\beta\}_{\beta \in \mathcal{A}_V}$ be the standard skew-symmetric Kraus operators for $\Lambda_U$ and $\Lambda_V$, normalized by

$$\langle L_\alpha, L_{\alpha'}\rangle_{\mathrm{HS}} = 2\delta_{\alpha\alpha'}, \qquad \langle M_\beta, M_{\beta'}\rangle_{\mathrm{HS}} = 2\delta_{\beta\beta'}.$$

Put

$$\mathcal{I}_r := \{(i,j,\alpha,\beta) : 1 \le i < j \le r,\ \alpha \in \mathcal{A}_U,\ \beta \in \mathcal{A}_V\}.$$

Define $\mathcal{T} : \ell_2(\mathcal{I}_r) \to U \otimes V \otimes Q$ by

$$\mathcal{T}c := \sum_{1 \le i<j \le r} \sum_{\alpha,\beta} c_{\alpha\beta}^{(ij)} (L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}. \tag{23}$$

Thus the column indexed by $(i,j,\alpha,\beta) \in \mathcal{I}_r$ is $(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}$. Using the Kraus form of $\Phi$ and the rank-one expansion of $P_-$ therefore gives

$$\begin{aligned} \Phi(P_-) &= \sum_{i<j} \sum_{\alpha,\beta} |(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}\rangle\langle (L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}| \\ &= \mathcal{T}\mathcal{T}^*. \end{aligned} \tag{24}$$

For each edge $(i,j)$ of the complete graph $K_r$, define

$$K_c^{(ij)} := \sum_{\alpha,\beta} c_{\alpha\beta}^{(ij)} L_\alpha \otimes M_\beta.$$

Then $K_c^{(ij)}$ lies in the linear span of

$$\{L \otimes M : L \in \mathfrak{so}(U),\ M \in \mathfrak{so}(V)\},$$

and orthogonality of the Kraus operators gives

$$\begin{aligned}\|K_c^{(ij)}\|_2^2 &= \sum_{\alpha,\beta}|c_{\alpha\beta}^{(ij)}|^2\|L_\alpha\otimes M_\beta\|_2^2 \\ &= 4\|c^{(ij)}\|_2^2.\end{aligned}\tag{25}$$

Grouping the output of $\mathcal{T}$ according to the basis $\{q_k\}$ gives

$$\mathcal{T}c = \frac{1}{\sqrt{2}}\sum_{k=1}^{r}\left(\sum_{i<k}K_c^{(ik)}\overline{e}_i - \sum_{k<j}K_c^{(kj)}\overline{e}_j\right)\otimes q_k. \tag{26}$$

At each vertex $k$, there are $r-1$ incident terms. Hence Cauchy–Schwarz yields

$$\begin{aligned}\|\mathcal{T}c\|^2 &\le \frac{r-1}{2}\sum_{k=1}^{r}\left(\sum_{i<k}\|K_c^{(ik)}\overline{e}_i\|^2 + \sum_{k<j}\|K_c^{(kj)}\overline{e}_j\|^2\right) \\ &= \frac{r-1}{2}\sum_{1\le i<j\le r}\left(\|K_c^{(ij)}\overline{e}_i\|^2 + \|K_c^{(ij)}\overline{e}_j\|^2\right).\end{aligned}\tag{27}$$

Each parenthesized expression is bounded by $\beta_2(\Phi)$ directly, without Equation (25) and without any property of $\mathfrak{so}(U)\otimes\mathfrak{so}(V)$. Write $\{A_a\}$ for the Kraus family of $\Phi$ on $\mathcal{L}(U\otimes V)$, here $A_{\alpha\beta} = L_\alpha\otimes M_\beta$; fix an edge and put $K := K_c^{(ij)}$, $c := c^{(ij)}$, so that $K = \sum_a c_a A_a$. Let $x, y$ be an orthonormal pair and $P := |x\rangle\langle x| + |y\rangle\langle y|$, so that $A := \|KP\|_2^2 = \|Kx\|^2 + \|Ky\|^2$ and $\operatorname{rank}(KP) \le 2$. Because $P$ is a self-adjoint idempotent, $\langle K, KP\rangle_{\mathrm{HS}} = A$; expanding $K$ along the Kraus family and applying Cauchy–Schwarz on the Kraus index,

$$A = \sum_a \overline{c_a}\,\langle A_a, KP\rangle_{\mathrm{HS}} \le \|c\|_2\Big(\sum_a|\langle A_a, KP\rangle_{\mathrm{HS}}|^2\Big)^{1/2} = \|c\|_2\,\langle \operatorname{vec}(KP), J(\Phi)\operatorname{vec}(KP)\rangle^{1/2} \le \|c\|_2\big(2\beta_2(\Phi)A\big)^{1/2}, \tag{28}$$

the middle equality by Definition 2.3 and the last step because $\operatorname{rank}(KP)\le 2$. If $A = 0$ there is nothing to prove; otherwise one factor of $A$ cancels, giving

$$\|Kx\|^2 + \|Ky\|^2 \le 2\beta_2(\Phi)\|c\|_2^2. \tag{29}$$

Neither minimality nor orthogonality of $\{A_a\}$ is used; the projection $Q_-$ enters only through the number $\beta_2(\Phi)$, and $\Pi K = K$ is replaced by membership of $K$ in the Kraus span. Substituting Equation (29) into Equation (27),

$$\begin{aligned}\|\mathcal{T}c\|^2 &\le \frac{r-1}{2}\cdot 2\beta_2(\Phi)\sum_{i<j}\|c^{(ij)}\|_2^2 \\ &= (r-1)\beta_2(\Phi)\|c\|_2^2.\end{aligned}\tag{30}$$

For $\Phi = \Lambda_U\otimes\Lambda_V$ this is $(r-1)\|c\|_2^2$ by Equation (11), and the estimate is attained already when $r=2$ and $\dim U = \dim V = 2$. Then every vertex has degree one and $\mathfrak{so}(2)\otimes\mathfrak{so}(2)$ is one-dimensional; its nonzero elements are scalar multiples of a unitary, so every orthonormal pair saturates Equation (4). There is no slack anywhere in that $r=2$ chain.

Thus

$$\mathcal{T}\mathcal{T}^* \preceq (r-1)\beta_2(\Phi)\, I_{UVQ}.$$

It remains to use the orthogonality of the range of $\mathcal{T}$ to $\delta_e$. Since each tensor product of two skew-symmetric operators is symmetric,

$$K^T = K \qquad \text{for every } K \text{ in the above linear span.}$$

Consequently,

$$\langle \delta_e, \mathcal{T}c\rangle = \frac{1}{\sqrt{2}}\sum_{i<j}\left(\bar{e}_j^T K_c^{(ij)}\bar{e}_i - \bar{e}_i^T K_c^{(ij)}\bar{e}_j\right) = 0. \tag{31}$$

Hence

$$\operatorname{ran}\mathcal{T} \subseteq \delta_e^{\perp}, \qquad P_{\delta_e^{\perp}} = I_{UVQ} - \Pi_e.$$

Since Equation (30) says that $\|\mathcal{T}\|^2 \le (r-1)\beta_2(\Phi)$, for every vector $x$,

$$\begin{aligned}\langle x, \mathcal{T}\mathcal{T}^* x\rangle &= \langle P_{\delta_e^{\perp}} x, \mathcal{T}\mathcal{T}^* P_{\delta_e^{\perp}} x\rangle \\ &\le (r-1)\beta_2(\Phi)\|P_{\delta_e^{\perp}} x\|^2.\end{aligned}$$

Equivalently,

$$\mathcal{T}\mathcal{T}^* \preceq (r-1)\beta_2(\Phi)(I_{UVQ} - \Pi_e).$$

This is Equation (12), and Equation (11) specializes it to Equation (22), whence $H_r^{(0)} \succeq 0$ by Equation (21). □

*Proof of Theorem 2.4.* The argument just given uses $\Phi$ only through three facts: that it is a Kraus sum, so that $\Phi(P_+) \succeq 0$ and Equation (24) holds; that its Kraus operators are transpose-symmetric, which is what makes Equation (31) vanish; and Equation (28), which uses only $\beta_2(\Phi)$ and membership of each $K_c^{(ij)}$ in the Kraus span. It therefore proves Equation (12) verbatim for any such $\Phi$.

For Equation (13), $P_+ - P_- = 2\rho_0^{T_{UV}}$ by Equation (16) and $\Phi(P_+) \succeq 0$, so $2\Phi(\rho_0^{T_{UV}}) \succeq -\Phi(P_-)$; Equation (12) bounds the right-hand side. Finally Equation (14) together with Equation (33) identifies $\Psi_{r,\Phi}$ with the map whose value at $\rho_0$ is the left-hand side of Equation (13), and Equation (40) supplies, for each $C$ of rank at most $r$, the orthonormal frame and the test vector at which to evaluate it; this is the $r$-positivity assertion. □

*Remark* 2.6 (Transpose parity, and where the trace term comes from). Since $\operatorname{vec}(K^T) = F\operatorname{vec}(K)$ for the flip $F$ on the two vectorization slots, transposition is an involution of the Kraus space and grades it by the flip eigenvalue: $\wedge^2 H$ has degree 1 and $\mathrm{Sym}^2 H$ degree 0. Kraus spaces multiply under $\otimes$ and the degree is additive, so the grading is by $\mathbb{Z}/2$. In this language:

> The hypothesis of Equation (31) is that the Kraus space has degree 0; each $\mathfrak{so}$ factor has degree 1; hence exactly the products of an *even* number of skew factors qualify.

Concretely, every $K$ in the span of $\{L \otimes M\}$ satisfies $K^T = K$, which is what makes the two terms of Equation (31) cancel. In odd degree $K^T = -K$, the two terms add, and $\operatorname{ran}\mathcal{T}$ does not lie in $\delta_e^{\perp}$; the argument then yields only $\mathcal{T}\mathcal{T}^* \preceq (r-1)I$ in place of $(r-1)(I - \Pi_e)$, giving $rI + \rho_0 - \iota_{UQ}(\rho_{UQ}^0) - \iota_{VQ}(\rho_{VQ}^0) \succeq 0$ and hence the coefficient 1 rather than the sharp $1/r$ in front of $|\operatorname{Tr} C|^2$.

So the $\frac{1}{r}|\operatorname{Tr} C|^2$ term of Theorem 1.1 is exactly the contribution of the single protected direction $\delta_e$, and the grading is what protects it: the generalizations that keep the sharp trace coefficient are the even-degree ones.

*Remark* 2.7 (The marginal inequality as an $r$-positivity statement). Write $\Delta := \eta \circ \varepsilon$ for discard-then-reprepare, where $\varepsilon = \operatorname{Tr}$ and $\eta(\lambda) = \lambda I$, so $\Delta(X) = \operatorname{Tr}(X)I$; let $\tau$ denote transposition; and recall $\mathcal{R}_\alpha(X) = \operatorname{Tr}(X)I + \alpha X$ from Corollary 4.6. Put

$$S := \mathcal{R}_{-1} = \Delta - \mathrm{id}, \qquad t := \frac{r-1}{r}, \qquad \text{so that} \qquad \mathcal{R}_{-1/r} = S + t\,\mathrm{id}.$$

Two facts suffice. First, $\Delta$ is monoidal: discarding a composite system and repreparing it is discarding and repreparing each factor, so $\Delta_{U\otimes V} = \Delta_U \otimes \Delta_V$; and $\tau$ is monoidal too. Since

$\Lambda(Y) = \mathrm{Tr}(Y)I - Y^T$ gives $\Lambda \circ \tau = \Delta - \mathrm{id} = S$ on a single factor,

$$(\Lambda_U \otimes \Lambda_V) \circ \tau = (\Lambda_U \circ \tau_U) \otimes (\Lambda_V \circ \tau_V) = S_U \otimes S_V. \tag{32}$$

Second, $\{\mathrm{id}, \Delta\}$ spans a two-dimensional commutative algebra, and the rest is a binomial expansion in it: using $\Delta = S + \mathrm{id}$ on each factor,

$$\begin{aligned} S \otimes S + (r-1)\,\Delta_U \otimes \Delta_V - t\,\mathrm{id} &= S \otimes S + (r-1)(S + \mathrm{id}) \otimes (S + \mathrm{id}) - t\,\mathrm{id} \\ &= r\,S \otimes S + (r-1)(S \otimes \mathrm{id} + \mathrm{id} \otimes S) + \tfrac{(r-1)^2}{r}\,\mathrm{id} \\ &= r\big[S \otimes S + t(S \otimes \mathrm{id} + \mathrm{id} \otimes S) + t^2\,\mathrm{id}\big], \end{aligned}$$

by $rt = r - 1$ and $rt^2 = (r-1)^2/r$. The bracket is $(S + t\,\mathrm{id})^{\otimes 2}$, so for every $r \geq 1$

$$(\Lambda_U \otimes \Lambda_V) \circ \tau + (r-1)\left(\Delta - \tfrac{1}{r}\mathrm{id}\right) = r\,\mathcal{R}^U_{-1/r} \otimes \mathcal{R}^V_{-1/r}. \tag{33}$$

Now $\mathrm{Tr}_{UV}\,\rho_0 = I_Q$ because the $e_i$ are orthonormal, and $\rho_0 = r\Pi_e$, so $I - \Pi_e = \big[(\Delta - \frac{1}{r}\mathrm{id}) \otimes \mathrm{id}_Q\big](\rho_0)$. Combining this with Equation (20) and Equation (33) rewrites Equation (14) as

$$H_r^{(0)} = \big[(r\,\mathcal{R}^U_{-1/r} \otimes \mathcal{R}^V_{-1/r}) \otimes \mathrm{id}_Q\big](\rho_0).$$

Thus Proposition 2.5 states precisely that $\mathcal{R}_{-1/r} \otimes \mathcal{R}_{-1/r}$ is $r$-positive. The passage from orthonormal frames $e_1, \dots, e_r$ to arbitrary vectors of Schmidt rank at most $r$ is the range factorization Equation (40), which produces the frame $\{e_i\}$ and the test vector $\delta_d$ in one step; no filtering or limiting argument is involved. Equivalently, this is the sufficiency half of Corollary 4.9 at $a = b = 1/r$, available here, before Theorem 1.1.

*Remark* 2.8 (Why the correction term has this shape). $\mathcal{L}(H) = \mathbb{C}I \oplus \{\text{traceless}\}$ is a decomposition into inequivalent $\mathrm{U}(d)$-representations, so the commutant — the $\mathrm{U}(d)$-covariant superoperators on $\mathcal{L}(H)$ — is exactly $\mathrm{span}\{\mathrm{id}, \Delta\}$, of dimension two, with $\mathcal{R}_{-a}$ acting as $d - a$ on $\mathbb{C}I$ and as $-a$ on the traceless part. A covariant correction to $\Phi \circ \tau$ therefore lies in a one-parameter pencil once its scale is fixed, and $1/r$ is the single free coefficient; Remark 2.6 identifies what pins it, namely the protected direction $\delta_e$. The correction term is not a choice.

The collapse in Equation (33) is specific to two factors. For $n$ factors monoidality gives $\Delta = \sum_{T \subseteq [n]} \bigotimes_{i \in T} S_i$, and matching against $c \bigotimes (S + t\,\mathrm{id})$ forces $r - 1 = r\,t^{n-|T|}$ for every $T$ with $|T| < n$. At $|T| = n - 1$ this reads $r - 1 = rt$, which holds; for $n = 2$ the $-t\,\mathrm{id}$ term repairs $|T| = 0$; but for $n \geq 3$ the equation at $|T| = n - 2$ is $r - 1 = (r-1)^2/r$, i.e. $r = 1$. So the even-partite generalizations of Remark 2.6 are genuine inequalities and cannot be recast as a map identity of the form $\bigotimes \mathcal{R}_{-a}$.

## 3. Proof of the Rank-$r$ Partial-Trace Inequality

**Lemma 3.1** (Partial-trace contraction identities)**.** *Let $e_1, \dots, e_s \in U \otimes V$ be orthonormal, let $d_1, \dots, d_s \in U \otimes V$, and let $q_1, \dots, q_s$ be an orthonormal basis of $Q = \mathbb{C}^s$. Define*

$$C := \sum_{i=1}^{s} |e_i\rangle\langle d_i|, \qquad \delta_e := \sum_{i=1}^{s} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|,$$

*and*

$$\delta_d := \sum_{i=1}^{s} d_i \otimes q_i, \qquad \rho^0_{UQ} := \mathrm{Tr}_V\,\rho_0, \qquad \rho^0_{VQ} := \mathrm{Tr}_U\,\rho_0.$$

*Then*

$$\langle \delta_d, \delta_d \rangle = \|C\|_2^2, \tag{34}$$

$$\langle \delta_d, \rho_0 \delta_d \rangle = |\mathrm{Tr}\,C|^2, \tag{35}$$

$$\langle \delta_d, \iota_{UQ}(\rho^0_{UQ}) \delta_d \rangle = \|\mathrm{Tr}_U\,C\|_2^2, \tag{36}$$

$$\langle \delta_d, \iota_{VQ}(\rho^0_{VQ}) \delta_d \rangle = \|\mathrm{Tr}_V\,C\|_2^2. \tag{37}$$

*Proof.* The first identity follows from the orthonormality of the $e_i$:

$$\|C\|_2^2 = \sum_i \|d_i\|^2 = \langle \delta_d, \delta_d \rangle.$$

Since $\operatorname{Tr} C = \sum_i \langle d_i, e_i \rangle$ under the stated inner-product convention,

$$\langle \delta_e, \delta_d \rangle = \sum_i \langle e_i, d_i \rangle = \overline{\operatorname{Tr} C},$$

which proves Equation (35).

For the first marginal contraction,

$$\rho^0_{UQ} = \sum_{i,j} \operatorname{Tr}_V |e_i\rangle\langle e_j| \otimes |q_i\rangle\langle q_j|,$$

and hence

$$\langle \delta_d, \iota_{UQ}(\rho^0_{UQ})\delta_d \rangle = \sum_{i,j} \langle d_i, (\operatorname{Tr}_V |e_i\rangle\langle e_j| \otimes I_V) d_j \rangle. \tag{38}$$

Choose product bases $\{u_a\}$ of $U$ and $\{v_b\}$ of $V$, and write

$$e_i = \sum_{a,b} e_{i,ab} u_a \otimes v_b, \qquad d_i = \sum_{a,b} d_{i,ab} u_a \otimes v_b.$$

Then

$$(\operatorname{Tr}_U C)_{b,c} = \sum_{i,a} e_{i,ab} \overline{d_{i,ac}},$$

so

$$\|\operatorname{Tr}_U C\|_2^2 = \sum_{i,j} \sum_{a,a'} \sum_{b,c} e_{i,ab} \overline{d_{i,ac}}\, \overline{e_{j,a'b}} d_{j,a'c}. \tag{39}$$

Expanding the right-hand side of Equation (38) gives, explicitly,

$$\begin{aligned} &\sum_{i,j} \langle d_i, (\operatorname{Tr}_V |e_i\rangle\langle e_j| \otimes I_V) d_j \rangle \\ &\qquad = \sum_{i,j} \sum_{a,a'} \sum_{b,c} e_{i,ab} \overline{d_{i,ac}}\, \overline{e_{j,a'b}} d_{j,a'c}, \end{aligned}$$

the same quadruple sum as in Equation (39). This proves Equation (36). Interchanging $U$ and $V$ proves Equation (37). $\square$

*Proof of Theorem 1.1.* The case $C = 0$ is immediate. Suppose $C \neq 0$, and let

$$r_0 := \operatorname{rank} C.$$

Choose an orthonormal basis $\{e_i\}_{i=1}^{r_0}$ of $\operatorname{ran} C$ and set

$$d_i := C^* e_i.$$

Then

$$C = \sum_{i=1}^{r_0} |e_i\rangle\langle d_i|, \qquad \sum_{i=1}^{r_0} \|d_i\|^2 = \|C\|_2^2. \tag{40}$$

Since $\operatorname{rank} C \le r$, one has $1 \le r_0 \le r$.

Let $Q = \mathbb{C}^{r_0}$, with orthonormal basis $\{q_i\}_{i=1}^{r_0}$, and define

$$\delta_e := \sum_{i=1}^{r_0} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle \delta_e|, \qquad \Pi_e := \frac{1}{r_0}\rho_0, \qquad \delta_d := \sum_{i=1}^{r_0} d_i \otimes q_i.$$

Write $\rho^0_{UQ} = \operatorname{Tr}_V \rho_0$ and $\rho^0_{VQ} = \operatorname{Tr}_U \rho_0$. Applying Proposition 2.5 with $r = r_0$ gives $H^{(0)}_{r_0} \succeq 0$. Therefore,

$$0 \le \langle \delta_d, H^{(0)}_{r_0} \delta_d \rangle$$

$$= r_0\langle\delta_d,\delta_d\rangle + \frac{1}{r_0}\langle\delta_d,\rho_0\delta_d\rangle - \langle\delta_d,\iota_{UQ}(\rho^0_{UQ})\delta_d\rangle - \langle\delta_d,\iota_{VQ}(\rho^0_{VQ})\delta_d\rangle. \tag{41}$$

By Lemma 3.1, this is precisely

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le r_0\|C\|_2^2 + \frac{1}{r_0}|\operatorname{Tr} C|^2. \tag{42}$$

It remains to pass from $r_0$ to $r$. From Equation (40) and $\operatorname{Tr} C = \sum_i \langle d_i, e_i\rangle$,

$$\begin{aligned}|\operatorname{Tr} C|^2 &\le \left(\sum_{i=1}^{r_0}\|d_i\|\right)^2\\ &\le r_0\sum_{i=1}^{r_0}\|d_i\|^2 = r_0\|C\|_2^2.\end{aligned}$$

The passage from $r_0$ to $r$ is then a single identity: for all $r \ge r_0 \ge 1$,

$$r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \left(r_0\|C\|_2^2 + \frac{1}{r_0}|\operatorname{Tr} C|^2\right) = (r - r_0)\left(\|C\|_2^2 - \frac{|\operatorname{Tr} C|^2}{rr_0}\right). \tag{43}$$

By the trace–rank bound the second factor is at least $\left(1-\frac{1}{r}\right)\|C\|_2^2 \ge 0$, and the first is nonnegative, so the right-hand side is nonnegative. Combining this with Equation (42) proves Equation (3).

Both factors are moreover *strictly* positive when $C \ne 0$ and $r_0 < r$: the first is at least 1, and $r_0 < r$ forces $r \ge 2$, so $1 - \frac{1}{r} \ge \frac{1}{2}$ while $\|C\|_2^2 > 0$. Hence the inequality of Equation (3) is strict whenever $\operatorname{rank} C < r$ and $C \ne 0$, which is the exact-rank assertion of Remark 1.2. □

*Remark* 3.2 (The bound at general $\beta_2$). Every constant in Sections 2 and 3 is affine in $\beta_2(\Phi)$, so contracting Equation (13) against $\delta_d$ exactly as above gives, for any $\Phi$ admissible in Theorem 2.4 and any nonzero $C$ with $r = \operatorname{rank} C$,

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le \big[r + (\beta_2 - 1)(r-1)\big]\|C\|_2^2 + \big[1 - (\beta_2 - 1)(r-1)\big]\frac{1}{r}|\operatorname{Tr} C|^2, \qquad \beta_2 := \beta_2(\Phi). \tag{44}$$

At $\beta_2 = 1$ this is Equation (2). (In terms of a two-vector action bound $\|Kx\|^2 + \|Ky\|^2 \le c\|K\|_2^2$ on a subspace of symmetric operators, $\beta_2 = 2c$; reading the family in $\beta_2$ rather than in $c$ removes the dependence on how the subspace is normalized, since $\beta_2$ is an invariant of the map.)

For $\Phi = \Lambda_U \otimes \Lambda_V$ the value $\beta_2 = 1$ cannot be improved: as noted after Equation (30) it is already attained at $\dim U = \dim V = 2$, where $\mathfrak{so}(2)\otimes\mathfrak{so}(2)$ is one-dimensional and every orthonormal pair saturates Equation (4). The extremizer of Remark 1.2 forces $\beta_2 \ge 1$ there unconditionally, so Equation (44) is never stronger than Theorem 1.1 on that space. Its interest is that $\beta_2$ is the *only* input: any completely positive map with transpose-symmetric Kraus operators yields a rank-$r$ partial-trace inequality with the constants displayed, and by the equivalence in Remark 2.2 at $p = 2$ the constant $\beta_2$ is optimal for each such map rather than merely sufficient.

*Remark* 3.3 (Two limits of the method). Theorem 2.4 does not extend in either of the two directions one would try first.

*Schmidt rank $p$.* Equation (28) generalizes at once: a rank-$p$ compression $P$ and the $S(p)$-norm of $J(\Phi)$ give $\sum_{i\le p}\|Kx_i\|^2 \le p\,\beta_p(\Phi)\|c\|_2^2$ for any orthonormal $p$-tuple. The reduction does not. The pair structure of Section 3 comes from the partial transpose and not from the Schmidt rank: $P_\pm$ are the $\pm1$ eigenspaces of the flip that $\rho_0^{T_{UV}}$ decomposes into, and $\eta_{ij}$, $\zeta_{ij}$, and the edge set of $K_r$ are pair-indexed downstream of that. A $p$-fold analogue would have to decompose the symmetric group $S_p$, which is a different problem. We record it as open.

*Higher copies.* By Remark 2.6, $\Phi = \Lambda_{U_1}\otimes\cdots\otimes\Lambda_{U_{2m}}$ has degree 0 and so satisfies the hypothesis of Theorem 2.4 for every $m$; expanding $\Phi\circ\tau$ then gives, for $\operatorname{rank} C \le r$,

$$\sum_{S\subseteq[2m]}(-1)^{|S|}\|\operatorname{Tr}_S C\|_2^2 + (r-1)\beta_2(\Phi)\left(\|C\|_2^2 - \tfrac{1}{r}|\operatorname{Tr} C|^2\right) \ge 0.$$

At $m = 1$ this is Equation (2). What is missing is the constant: $\beta_2(\Lambda^{\otimes 2m})$ for $m \geq 2$ is a Ky-Fan-2 concentration constant for the $2m$-fold antisymmetric projection, and we do not know its value. So the family *reduces* to one rank-two Choi $S(2)$-norm per $m$; it does not follow from Lemma 2.1. By Remark 2.8 it also cannot be recast as a map identity.

## 4. Applications and Corollaries

This section collects consequences of the results proved above: the exact symmetric score curve, an asymmetric tensor-product score theorem, block positivity and positive maps, quantitative Schmidt-number separation, Ky Fan bounds for Kronecker sums, and the exact Schmidt number of a product of antisymmetric projector states. None of them is part of the machine-checked development.

They do not all rest on the same input. Proposition 4.1, Theorem 4.8, and Corollary 4.11 use Theorem 1.1 at general $r$, and are the results for which the extension beyond normal matrices is what is needed; Corollaries 4.5, 4.6 and 4.9 restate them through the Choi–Jamiołkowski correspondence. Corollaries 4.12 and 4.13, by contrast, rest on Lemma 2.1 alone and do not use Theorem 1.1. Appendix C collects two further consequences that use nothing about the witnesses constructed here beyond $r$-block positivity.

### 4.1. Two-copy block positivity.

Fix $d \geq 2$ and $\alpha \in \mathbb{R}$. Let

$$\rho_\alpha := I + \alpha F$$

be the unnormalized Werner operator on $\mathbb{C}^d \otimes \mathbb{C}^d$; for $\alpha \in [-1, 1]$, it is proportional to the standard Werner state [9]. Its partial transpose is

$$\rho_\alpha^\Gamma = I + \alpha|\Omega\rangle\langle\Omega|, \qquad \Omega := \sum_{i=1}^{d} e_i \otimes e_i.$$

The case $\alpha \geq 0$ is immediate because

$$I + \alpha|\Omega\rangle\langle\Omega| \succeq 0.$$

For $\alpha \leq 0$ and $C \in M_{d^2}(\mathbb{C})$, define

$$q^{(2)}(\alpha, C) := \|C\|_2^2 + \alpha\left(\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2\right) + \alpha^2|\operatorname{Tr} C|^2. \tag{45}$$

The first argument is the Werner parameter of $\rho_\alpha$; only $\alpha \leq 0$ is at issue, and it is convenient to write $\alpha = -t$ with $t \geq 0$, so that

$$q^{(2)}(-t, C) = \|C\|_2^2 - t\left(\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2\right) + t^2|\operatorname{Tr} C|^2.$$

An operator is called $r$-block-positive if its expectation is nonnegative on every vector of Schmidt rank at most $r$. With the vectorization convention fixed in the introduction and the regrouping $(A_1A_2) : (B_1B_2)$, one has

$$\begin{aligned}&\left\langle \operatorname{vec}(C), (\rho_\alpha^\Gamma)^{\otimes 2} \operatorname{vec}(C)\right\rangle \\ &= q^{(2)}(\alpha, C),\end{aligned} \tag{46}$$

which is what the first argument of $q^{(2)}$ records. Moreover,

$$\operatorname{SR}\big(\operatorname{vec}(C)\big) = \operatorname{rank} C$$

across the same output–input regrouping. Thus $r$-block positivity of $(\rho_{-t}^\Gamma)^{\otimes 2}$ is equivalent to $q^{(2)}(-t, C) \geq 0$ for every matrix $C$ of rank at most $r$. For every $C$ of rank at most $r$, Equation (3) implies

$$q^{(2)}(-t, C) \geq (1 - rt)\|C\|_2^2 + \left(t^2 - \frac{t}{r}\right)|\operatorname{Tr} C|^2. \tag{47}$$

If $0 \le t \le 1/r$, then $t^2 - t/r \le 0$. Using

$$|\operatorname{Tr} C|^2 \le r\|C\|_2^2$$

in the appropriate direction gives

$$\begin{aligned} q^{(2)}(-t, C) &\ge \left(1 - rt + rt^2 - t\right) \|C\|_2^2 \\ &= (1-t)(1-rt)\|C\|_2^2 \ge 0. \end{aligned} \tag{48}$$

**Proposition 4.1** (Exact rank-constrained two-copy score curve)**.** *Let $d \ge 2$ and $1 \le r \le d$. For $t \ge 0$, define*

$$\mu_r(t) := \inf_{\substack{C \in M_{d^2}(\mathbb{C}) \setminus \{0\} \\ \operatorname{rank} C \le r}} \frac{q^{(2)}(-t, C)}{\|C\|_2^2}.$$

*Then*

$$\mu_r(t) = \begin{cases} (1-t)(1-rt), & 0 \le t \le 1/r, \\ 1 - rt, & t \ge 1/r. \end{cases} \tag{49}$$

*Both branches are attained by rank-$r$ matrices.*

*Proof.* For $C \ne 0$ of rank at most $r$, set

$$z := \frac{|\operatorname{Tr} C|^2}{\|C\|_2^2}.$$

The trace–rank estimate gives $0 \le z \le r$. Dividing Equation (47) by $\|C\|_2^2$ yields

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \ge 1 - rt + \left(t^2 - \frac{t}{r}\right) z. \tag{50}$$

If $0 \le t \le 1/r$, the coefficient of $z$ is nonpositive, so Equation (50) and $z \le r$ give

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \ge 1 - rt + r\left(t^2 - \frac{t}{r}\right) = (1-t)(1-rt).$$

Choose a rank-$r$ projection $P_r \in M_d(\mathbb{C})$ and a unit vector $v \in \mathbb{C}^d$. For

$$C = P_r \otimes |v\rangle\langle v|$$

one has

$$\begin{aligned} \|C\|_2^2 = r, \qquad |\operatorname{Tr} C|^2 = r^2, \\ \|\operatorname{Tr}_1 C\|_2^2 = r^2, \quad \|\operatorname{Tr}_2 C\|_2^2 = r, \end{aligned}$$

and hence equality holds.

If $t \ge 1/r$, the coefficient of $z$ in Equation (50) is nonnegative, so

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \ge 1 - rt.$$

Choose orthogonal unit vectors $v, w \in \mathbb{C}^d$ and set

$$C = P_r \otimes |v\rangle\langle w|.$$

Then

$$\operatorname{Tr} C = 0, \qquad \|\operatorname{Tr}_1 C\|_2^2 = r^2, \qquad \operatorname{Tr}_2 C = 0,$$

so equality again holds. ∎

*Remark* 4.2 (Why the range $r \le d$ is essential). The extremizers in Proposition 4.1 require a rank-$r$ projection in one $d$-dimensional tensor factor. The score formula does not in general extend to $r > d$ by replacing $r$ with $\min\{r, d\}$. For example, when the rank bound is $r = d^2$, no rank constraint remains. If $0 \le t \le 1/d$, the smallest eigenvalue of

$$\left(\rho_{-t}^{\Gamma}\right)^{\otimes 2}$$

is

$$(1 - dt)^2.$$

By Equation (46), the displayed infimum is the smallest eigenvalue, and therefore

$$\inf_{C \ne 0} \frac{q^{(2)}(-t, C)}{\|C\|_2^2} = (1 - dt)^2.$$

This generally differs from $(1 - t)(1 - dt)$, the expression obtained by formally substituting $d$ for $r$ in the first branch of Equation (49).

**Corollary 4.3** (An explicit adjacent-Schmidt-rank violation)**.** *Let $1 \le r < d$ and put*

$$Z_{r,d} := \left(I - \frac{1}{r}|\Omega\rangle\langle\Omega|\right)^{\otimes 2}.$$

*Across the regrouped bipartition $(A_1A_2) : (B_1B_2)$,*

$$\langle\psi, Z_{r,d}\psi\rangle \ge 0 \qquad \text{whenever } \|\psi\| = 1 \text{ and } \mathrm{SR}(\psi) \le r.$$

*On the other hand, there is a unit vector $\psi_{r+1}$ of Schmidt rank $r + 1$ such that*

$$\langle\psi_{r+1}, Z_{r,d}\psi_{r+1}\rangle = -\frac{1}{r}. \tag{51}$$

*Proof.* The nonnegativity assertion is Corollary 4.5 at $t = 1/r$. Choose a rank-$(r + 1)$ projection $P_{r+1}$ and orthogonal unit vectors $v, w \in \mathbb{C}^d$, and define

$$C = P_{r+1} \otimes |v\rangle\langle w|, \qquad \psi_{r+1} := \frac{\mathrm{vec}(C)}{\sqrt{r+1}}.$$

Then $\mathrm{SR}(\psi_{r+1}) = \operatorname{rank} C = r + 1$, while

$$\frac{q^{(2)}(-1/r, C)}{\|C\|_2^2} = 1 - \frac{r+1}{r} = -\frac{1}{r}.$$

Now use Equation (46). ☐

*Remark* 4.4 (A strict signed separation). Let $0 < \varepsilon < 1/(r + 1)$ and set $t = (1 - \varepsilon)/r$. Then Proposition 4.1 gives

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \ge \varepsilon\left(1 - \frac{1-\varepsilon}{r}\right) > 0 \qquad (\operatorname{rank} C \le r),$$

whereas the rank-$(r + 1)$ matrix used in Corollary 4.3 has normalized score

$$1 - (r + 1)t = \frac{(r + 1)\varepsilon - 1}{r} < 0.$$

Thus the filter has a uniform positive margin on all normalized matrices of rank at most $r$, while the displayed rank-$(r + 1)$ matrix has negative score.

This yields the exact $r$-block-positivity threshold for the two-copy family $(\rho_\alpha^\Gamma)^{\otimes 2}$.

**Corollary 4.5** (Two-copy block positivity)**.** *Let $d \geq 2$ and $\alpha \in \mathbb{R}$. For every $1 \leq r \leq d^2$, the operator*

$$(\rho_\alpha^\Gamma)^{\otimes 2}$$

*is $r$-block-positive across the regrouped bipartition*

$$(A_1A_2) : (B_1B_2)$$

*if and only if*

$$\alpha \geq -\frac{1}{\min\{r, d\}}. \tag{52}$$

*Proof.* It remains only to consider $\alpha = -t < 0$.

Suppose first that $r \leq d$. If $t \leq 1/r$, then Equation (48) gives

$$q^{(2)}(-t, C) \geq 0$$

for every $C$ of rank at most $r$, proving $r$-block positivity.

For necessity in all cases, set

$$s := \min\{r, d\}.$$

Choose a rank-$s$ projection $P_s$ and orthogonal unit vectors $v, w \in \mathbb{C}^d$, and let

$$C = P_s \otimes |v\rangle\langle w|.$$

Then

$$\begin{aligned} \operatorname{rank} C &= s \leq r, \\ \operatorname{Tr} C &= 0, \\ \|C\|_2^2 &= s, \\ \|\operatorname{Tr}_1 C\|_2^2 &= s^2, \\ \|\operatorname{Tr}_2 C\|_2^2 &= 0. \end{aligned}$$

Hence

$$q^{(2)}(-t, C) = s(1 - st),$$

which is negative whenever $t > 1/s$. Thus $r$-block positivity forces

$$t \leq \frac{1}{\min\{r, d\}}.$$

Finally, suppose $r > d$. If $t \leq 1/d$, then the Choi operator of

$$\mathcal{R}_{-t}(X) := \operatorname{Tr}(X)I - tX$$

is

$$J(\mathcal{R}_{-t}) = I - t|\Omega\rangle\langle\Omega| \succeq 0,$$

because $\|\Omega\|^2 = d$ and the only potentially nonpositive eigenvalue is $1 - td$. Therefore $\mathcal{R}_{-t}$ is completely positive, so $\mathcal{R}_{-t}^{\otimes 2}$ is completely positive and hence $r$-positive for every $r$. This proves sufficiency when $r > d$. □

4.2. **$r$-positivity of the tensor square.** The preceding block-positivity statement has an equivalent map formulation through the Choi–Jamiołkowski correspondence: a map is $r$-positive if and only if its Choi operator is $r$-block-positive [1, 8]. With the output factors grouped against the input factors, the Choi operator of $\mathcal{R}_\alpha^{\otimes 2}$ is exactly $(\rho_\alpha^\Gamma)^{\otimes 2}$ across the bipartition $(A_1A_2):(B_1B_2)$ used above.

**Corollary 4.6** (Tensor-square stability)**.** *Let $d \geq 2$ and $\alpha \in \mathbb{R}$. Define*

$$\mathcal{R}_\alpha(X) := \operatorname{Tr}(X)I + \alpha X.$$

*For every $1 \leq r \leq d^2$, the tensor-square map*

$$\mathcal{R}_\alpha^{\otimes 2}$$

*is $r$-positive if and only if*

$$\alpha \geq -\frac{1}{\min\{r,d\}}.$$

4.3. **Exact asymmetric tensor-product theorem.** For $a, b \geq 0$, define

$$q_{a,b}(C) := \|C\|_2^2 - a\|\operatorname{Tr}_1 C\|_2^2 - b\|\operatorname{Tr}_2 C\|_2^2 + ab|\operatorname{Tr} C|^2. \tag{53}$$

Under the same vectorization and regrouping as above,

$$q_{a,b}(C) = \langle \operatorname{vec}(C), (I - a|\Omega\rangle\langle\Omega|) \otimes (I - b|\Omega\rangle\langle\Omega|) \operatorname{vec}(C)\rangle\,.$$

**Lemma 4.7** (One-sided rank bound)**.** *If $\operatorname{rank} C \leq r$, then*

$$\|\operatorname{Tr}_j C\|_2^2 \leq r\|C\|_2^2, \qquad j \in \{1,2\}. \tag{54}$$

*Proof.* Write $C = \sum_{i=1}^{r_0} |e_i\rangle\langle d_i|$ as in Equation (40), with $e_1, \ldots, e_{r_0}$ orthonormal and $r_0 = \operatorname{rank} C \leq r$. Identify a vector $x$ of the bipartite space with its coefficient matrix $X$ relative to the chosen product basis, so that $\|X\|_2 = \|x\|$. The two partial traces of a rank-one operator are then matrix products,

$$\operatorname{Tr}_2|x\rangle\langle y| = XY^*, \qquad \operatorname{Tr}_1|x\rangle\langle y| = \overline{X^*Y}, \tag{55}$$

both immediate from the definitions. The Hilbert–Schmidt norm is submultiplicative, so each summand obeys $\|\operatorname{Tr}_j|e_i\rangle\langle d_i|\|_2 \leq \|e_i\|\|d_i\| = \|d_i\|$. By the triangle inequality and then Cauchy–Schwarz,

$$\|\operatorname{Tr}_j C\|_2^2 \leq \Big(\sum_{i=1}^{r_0}\|d_i\|\Big)^2 \leq r_0\sum_{i=1}^{r_0}\|d_i\|^2 = r_0\|C\|_2^2 \leq r\|C\|_2^2,$$

the equality being Equation (40). $\square$

Recall that $\operatorname{Tr}_1$ means the trace *over* the first local factor, leaving an operator on the second, while $\operatorname{Tr}_2$ traces over the second and leaves an operator on the first. Thus the coefficient $a$ in Equation (53) multiplies the first-factor trace and $b$ multiplies the second-factor trace.

**Theorem 4.8** (Exact asymmetric tensor-product score)**.** *Let $d \geq 2$, $1 \leq r \leq d$, and $a, b \geq 0$. Put*

$$m := \max\{a,b\}, \qquad \ell := \min\{a,b\}.$$

*Then*

$$\inf_{\substack{C \in M_{d^2}(\mathbb{C})\setminus\{0\} \\ \operatorname{rank} C \leq r}} \frac{q_{a,b}(C)}{\|C\|_2^2} = \begin{cases} (1-\ell)(1-rm), & m \leq 1/r, \\ 1 - rm, & m \geq 1/r. \end{cases} \tag{56}$$

*Proof.* By interchanging the two tensor factors, assume $a = m$ and $b = \ell$. For $C \neq 0$ of rank at most $r$, write

$$x := \frac{\|\operatorname{Tr}_1 C\|_2^2}{\|C\|_2^2}, \qquad y := \frac{\|\operatorname{Tr}_2 C\|_2^2}{\|C\|_2^2}, \qquad z := \frac{|\operatorname{Tr} C|^2}{\|C\|_2^2}.$$

By Theorem 1.1 and Lemma 4.7,

$$x + y \leq r + \frac{z}{r}, \qquad x \leq r, \qquad 0 \leq z \leq r.$$

Consequently,

$$\begin{aligned} ax + by &= b(x+y) + (a-b)x \\ &\leq b\left(r + \frac{z}{r}\right) + (a-b)r = ar + \frac{b}{r}z, \end{aligned}$$

and hence

$$\frac{q_{a,b}(C)}{\|C\|_2^2} \geq 1 - ar + b\left(a - \frac{1}{r}\right)z. \tag{57}$$

If $a \leq 1/r$, minimize the right-hand side over $0 \leq z \leq r$ by taking $z = r$; this gives

$$1 - ar + br\left(a - \frac{1}{r}\right) = (1-b)(1-ar).$$

Equality is attained by $C = P_r \otimes |v\rangle\langle v|$. If $a \geq 1/r$, minimize by taking $z = 0$; this gives $1 - ar$, with equality attained by $C = P_r \otimes |v\rangle\langle w|$ for $v \perp w$. If $b > a$, interchange the two tensor factors in these constructions. □

**Corollary 4.9** (Asymmetric tensor-product $r$-positivity threshold)**.** *Let $d \geq 2$, $1 \leq r \leq d^2$, and $a, b \geq 0$. Then*

$$(I - a|\Omega\rangle\langle\Omega|) \otimes (I - b|\Omega\rangle\langle\Omega|)$$

*is $r$-block-positive across the regrouped bipartition if and only if*

$$\max\{a, b\} \leq \frac{1}{\min\{r, d\}}. \tag{58}$$

*Equivalently,*

$$\mathcal{R}_{-a} \otimes \mathcal{R}_{-b}$$

*is $r$-positive if and only if Equation (58) holds.*

*Proof.* The corner $a = b = 1/r$ of the sufficiency half is Remark 2.7, which derives it from Proposition 2.5 directly; it does not depend on this section. For the general case with $r \leq d$, the assertion follows from Theorem 4.8. (That route passes through Theorem 1.1, hence through Proposition 2.5; the remark shows the $a = b = 1/r$ corner without the detour. The other corners do need an argument rather than a citation: $\mathcal{R}_{-a} = \mathcal{R}_{-1/r} + (\frac{1}{r} - a)\mathrm{id}$, and the cross terms $\mathcal{R}_{-1/r} \otimes \mathrm{id}$ are not separately positive, so monotonicity in $a$ and $b$ is not immediate.) If $r > d$ and $a, b \leq 1/d$, both factors $I - a|\Omega\rangle\langle\Omega|$ and $I - b|\Omega\rangle\langle\Omega|$ are positive semidefinite, so their tensor product is positive semidefinite. Conversely, if, say, $a > 1/d$, choose

$$C = I_d \otimes |v\rangle\langle w|, \qquad v \perp w.$$

Then $\operatorname{rank} C = d \leq r$ and $q_{a,b}(C)/\|C\|_2^2 = 1 - ad < 0$. The case $b > 1/d$ follows by interchanging the tensor factors. The map statement follows from the Choi–Jamiołkowski correspondence. □

4.4. **Quantitative trace-norm separation from bounded Schmidt number.**

**Corollary 4.10** (Trace-norm separation certified by the two-copy witness)**.** *Let $1 \leq r < d$ and let $Z_{r,d}$ be as in Corollary 4.3. Put*

$$\gamma := \frac{d}{r} - 1 > 0, \qquad \Delta_{r,d} := \gamma + \max\{1, \gamma^2\}.$$

*Define the normalized Schmidt-number-$r$ state set*

$$\mathcal{S}_r := \{\sigma \succeq 0 : \operatorname{Tr}\sigma = 1,\ \mathrm{SN}(\sigma) \leq r\}.$$

*If $\rho$ is a state satisfying*

$$\operatorname{Tr}(Z_{r,d}\rho) = -\varepsilon < 0,$$

*then*

$$\inf_{\sigma\in\mathcal{S}_r} \|\rho-\sigma\|_1 \geq \frac{2\varepsilon}{\Delta_{r,d}}. \tag{59}$$

*Proof.* Since $Z_{r,d}$ is $r$-block-positive,

$$\operatorname{Tr}(Z_{r,d}\sigma) \geq 0 \qquad \text{for every } \sigma \in \mathcal{S}_r.$$

The operator $I - r^{-1}|\Omega\rangle\langle\Omega|$ has eigenvalues 1 and $1-d/r = -\gamma$. Hence $Z_{r,d}$ has eigenvalues

$$1, \qquad -\gamma, \qquad \gamma^2.$$

Thus its spectral diameter is $\Delta_{r,d}$.

Fix $\sigma \in \mathcal{S}_r$. Then

$$\operatorname{Tr}[Z_{r,d}(\sigma-\rho)] \geq \varepsilon.$$

Because $\operatorname{Tr}(\sigma-\rho)=0$, we may subtract from $Z_{r,d}$ the midpoint of its extreme eigenvalues. The resulting centered operator has operator norm $\Delta_{r,d}/2$. Hölder duality therefore gives

$$\varepsilon \leq \frac{\Delta_{r,d}}{2}\|\sigma-\rho\|_1.$$

Taking the infimum over $\sigma\in\mathcal{S}_r$ proves the claim. ∎

4.5. **Kronecker-sum Ky Fan bounds.** By dualizing the partial-trace inequality over rank-$k$ test operators, Theorem 1.1 yields a singular-value hierarchy for Kronecker sums.

**Corollary 4.11.** *Let $A, B \in M_d(\mathbb{C})$ be traceless, not necessarily Hermitian, operators, and define*

$$M := A\otimes I_d + I_d \otimes B.$$

*For every $1 \leq k \leq d^2$,*

$$\sum_{j=1}^{k} s_j(M)^2 \leq \min\left\{d, k+\max\left\{0, 1-\frac{2k}{d}\right\}\right\}\left(\|A\|_2^2+\|B\|_2^2\right). \tag{60}$$

*In particular, if $d \leq 2k$, then*

$$\sum_{j=1}^{k} s_j(M)^2 \leq k\left(\|A\|_2^2+\|B\|_2^2\right).$$

*Proof.* The Frobenius-dual form of Ky Fan duality gives

$$\left(\sum_{j=1}^{k} s_j(M)^2\right)^{1/2} = \max_{\substack{\operatorname{rank} C\leq k\\ \|C\|_2=1}} |\langle C, M\rangle_{\mathrm{HS}}|. \tag{61}$$

Since $A$ and $B$ are traceless,

$$\begin{aligned}\langle C, M\rangle_{\mathrm{HS}} = {}& \left\langle \operatorname{Tr}_2 C - \frac{\operatorname{Tr} C}{d} I, A\right\rangle_{\mathrm{HS}} \\ & + \left\langle \operatorname{Tr}_1 C - \frac{\operatorname{Tr} C}{d} I, B\right\rangle_{\mathrm{HS}}.\end{aligned} \tag{62}$$

By Cauchy–Schwarz,

$$\begin{aligned}|\langle C, M\rangle_{\mathrm{HS}}|^2 \leq {}& \left(\|A\|_2^2+\|B\|_2^2\right) \\ & \times\left(\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2 - \frac{2}{d}|\operatorname{Tr} C|^2\right).\end{aligned} \tag{63}$$

For a rank-at-most-$k$ operator $C$ with $\|C\|_2 = 1$, Equation (3) gives

$$\begin{aligned}&\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2 - \frac{2}{d}|\operatorname{Tr} C|^2 \\ &\leq k + \left(\frac{1}{k} - \frac{2}{d}\right)|\operatorname{Tr} C|^2. \end{aligned} \tag{64}$$

If $1/k - 2/d \leq 0$, the final term may be discarded. If $1/k - 2/d > 0$, then

$$|\operatorname{Tr} C|^2 \leq k\|C\|_2^2 = k.$$

Thus the right-hand side of Equation (64) is at most

$$k + \max\left\{0, 1 - \frac{2k}{d}\right\}.$$

Combining this with Equations (61) and (63) gives the second entry in the minimum in Equation (60). On the other hand, tracelessness implies

$$\|M\|_2^2 = d\left(\|A\|_2^2 + \|B\|_2^2\right),$$

which gives the first entry. This proves Equation (60). □

4.6. **Exact Schmidt number and witnesses.** For $m \geq 2$, let $\Pi_-^{(m)}$ denote the orthogonal projection onto the antisymmetric subspace of $\mathbb{C}^m \otimes \mathbb{C}^m$, and let

$$\omega_-^{(m)} := \frac{\Pi_-^{(m)}}{\binom{m}{2}}$$

denote the corresponding normalized state.

The argument below invokes Lemma 2.1 directly rather than Theorem 1.1: what it needs is the bound on the top two Schmidt coefficients of a double-skew operator, not the partial-trace inequality derived from it. That $\operatorname{SN}(\omega_-^{(m)}) = 2$ is classical; the content here is the value for the tensor product.

**Corollary 4.12** (Exact Schmidt number)**.** *With respect to the regrouped bipartition*

$$(A_1A_2) : (B_1B_2),$$

*one has, for every $m, n \geq 2$,*

$$\operatorname{SN}\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right) = 4. \tag{65}$$

*Proof.* Each standard antisymmetric basis vector has Schmidt rank 2, so $\omega_-^{(m)}$ admits a decomposition into pure states of Schmidt rank 2. Moreover, the antisymmetric subspace contains no nonzero product vector. Hence

$$\operatorname{SN}(\omega_-^{(m)}) = 2.$$

Tensoring Schmidt-rank-2 decompositions therefore gives

$$\operatorname{SN}\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right) \leq 4.$$

For the reverse inequality, let

$$Q := \Pi_-^{(m)} \otimes \Pi_-^{(n)},$$

where the first projection acts on $A_1B_1$ and the second on $A_2B_2$. Under inverse vectorization across $(A_1A_2) : (B_1B_2)$, every vector $\phi \in \operatorname{ran} Q$ has the form

$$\phi = \operatorname{vec}(K), \qquad K \in \mathfrak{so}(\mathbb{C}^m) \otimes \mathfrak{so}(\mathbb{C}^n).$$

Thus, for a unit vector $\phi \in \operatorname{ran} Q$, its Schmidt coefficients $s_1 \geq s_2 \geq s_3 \geq \cdots$ across the regrouped bipartition are the singular values of a Hilbert–Schmidt unit operator $K$ in the above double-skew space. The case $m \neq n$ is included by the zero-extension argument of Remark 2.2. By that lemma,

$$s_1^2 + s_2^2 \leq \frac{1}{2}.$$

Since $s_3 \leq s_2 \leq s_1$,

$$2s_3^2 \leq s_1^2 + s_2^2 \leq \frac{1}{2}.$$

Consequently,

$$s_1^2 + s_2^2 + s_3^2 \leq \frac{3}{4}.$$

For any orthogonal projection $Q$ and any $k$, projection duality gives

$$\begin{aligned} \sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq k}} \langle z, Qz\rangle &= \sup_{\substack{\phi\in\operatorname{ran} Q \\ \|\phi\|=1}} \sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq k}} |\langle z, \phi\rangle|^2 \\ &= \sup_{\substack{\phi\in\operatorname{ran} Q \\ \|\phi\|=1}} \sum_{j=1}^{k} s_j(\phi)^2; \end{aligned}$$

see [7]. Applying this identity with $k = 3$ gives

$$\sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq 3}} \langle z, Qz\rangle \leq \frac{3}{4}.$$

On the other hand,

$$\omega_-^{(m)} \otimes \omega_-^{(n)} = \frac{Q}{\operatorname{rank} Q},$$

and hence

$$\operatorname{Tr}\left[Q\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = 1.$$

If the state had Schmidt number at most 3, it would admit a convex decomposition into pure states of Schmidt rank at most 3 [8]; its expectation against $Q$ would then be at most $3/4$, a contradiction. Therefore its Schmidt number is at least 4, proving Equation (65). □

**Corollary 4.13** (Explicit Schmidt-number witnesses)**.** *Let*

$$Q = \Pi_-^{(m)} \otimes \Pi_-^{(n)}.$$

*Then*

$$W_2 := \frac{1}{2} I - Q$$

*satisfies*

$$\operatorname{Tr}(W_2\sigma) \geq 0 \qquad \textit{whenever } \mathrm{SN}(\sigma) \leq 2,$$

*and*

$$W_3 := \frac{3}{4} I - Q$$

*satisfies*

$$\operatorname{Tr}(W_3\sigma) \geq 0 \qquad \textit{whenever } \mathrm{SN}(\sigma) \leq 3.$$

*Thus a negative expectation of $W_2$ certifies Schmidt number at least 3, while a negative expectation of $W_3$ certifies Schmidt number at least 4. For the state in Corollary 4.12,*

$$\operatorname{Tr}\left[W_2\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = -\frac{1}{2},$$

$$\operatorname{Tr}\left[W_3\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = -\frac{1}{4}.$$

## Appendix A. Formal verification

The proof of Theorem 1.1, and of every result it depends on, has been formalized in Lean 4. The development is available at `https://github.com/elazarg/rank-r`. It uses the toolchain `leanprover/lean4:v4.32.0`, and Mathlib is its only direct dependency, pinned in `lake-manifest.json` at revision `81a5d257c8e410db227a6665ed08f64fea08e997` (tag `v4.32.0`); that manifest also pins every transitive dependency. After `lake exe cache get`, the single command `lake build RankR` compiles it. The sources contain no `sorry`, no `axiom` declaration, no `native_decide` and no `set_option`, and `#print axioms` applied to the main theorem reports only `propext`, `Classical.choice` and `Quot.sound`; the file `RankR/Axioms.lean` wraps that check in `#guard_msgs` for each headline theorem, so it is verified by the build rather than asserted here.

The formal counterpart of Theorem 1.1 is `rank_r_partial_trace`, and it has *no* hypotheses. The Autonne–Takagi factorization is itself proved, as `Matrix.exists_takagi`; `Takagi.lean` derives Lemma 2.1 from it, `Equivalence.lean` converts that into $\beta_2(\Lambda_U \otimes \Lambda_V) = 1$, and `Results.lean` composes the chain, so the development is proved end to end. The single quantitative input is that one number: the development's only route from Section 2 to Section 3 is Definition 2.3. Beyond Equation (3) itself the development covers the strict form of Equation (43), hence the assertion that for nonzero $C$ equality in Equation (3) forces $\operatorname{rank} C = r$; all of Remark 1.2 — the extremizer, its rank, the equality it attains, and the optimality of both coefficients; and Lemma 4.7.

Both directions of Remark 2.2 are formalized, so [4, Thm. 2.4] is machine-checked as well, as `fgpBound_holds`. Two things should still be stated precisely. First, the mathematics of Section 2 is theirs; the formal development follows their §2 and replaces their Theorem 2.4 by Autonne–Takagi, which removes their appeals to a max–min exchange and to [7], but it is not an independent derivation. Second, what is proved is `FGPBound` as Section 2 renders it, in the homogeneous form over $|u_1\rangle\langle v_1| + |u_2\rangle\langle v_2|$ rather than over unit vectors of Schmidt rank at most two; by Equation (1) these are the same statement, but that identification is a reading of the definitions rather than a theorem of the development.

The theorems `rank_r_partial_trace_of_FGP_square` and its variants are retained in the development as a bridge to the published estimate: they carry the hypothesis `FGPBound (Fin d) (Fin d)` together with $\operatorname{card} U \le d$ and $\operatorname{card} V \le d$, and record that Fu–Gao–Park's statement, in the square form they publish, yields the same conclusion. They are no longer on the critical path. The correspondence with the results of the main text is as follows.

| Manuscript | Lean |
|---|---|
| Theorem 1.1 | `rank_r_partial_trace` |
| Autonne–Takagi, [5, Cor. 4.4.4] | `Matrix.exists_takagi` |
| Lemma 2.1 | `doubleSkewBound_of_takagi` |
| Equation (6) | `hsInner_symOuter_flipU` |
| Equation (7) | `hsInner_flipU_pairMat_nonneg` |
| Equation (8) | `qform_Qm_vec_pairMat_le` |
| Equation (9) | `sq_add_sq_le_of_takagi` |
| weights in step 4 of Lemma 2.1 | `exists_pair_sum_weighted_le` |
| Remark 2.2, forward | `doubleSkewBound_of_FGP` |
| Remark 2.2, converse | `FGPBound_of_doubleSkewBoundQm` |
| [4, Thm. 2.4] | `fgpBound_holds` |
| zero extension in Remark 2.2 | `FGPBound_of_card_le` |
| Definition 2.3, $J(\Phi)$ | `choiOf` |
| Definition 2.3, $\beta_2$ | `ChoiTwoBound` |
| Equation (11) | `choiOf_skewKraus`, `choiTwoBound_skewKraus` |
| Equation (28), Equation (29) | `norm_sq_pair_le_of_choiTwoBound` |
| Equation (12) | `qform_krausQ_Pneg_le` |
| Equation (13) | `qform_krausQ_ptransposeUV_ge` |
| Equation (33) | `Psi_eq_tensor_square` |
| $\Delta_{U\otimes V} = \Delta_U \otimes \Delta_V$ | `RU_zero_RV_zero` |
| Equation (32) | `Lam4_transpose` |
| Proposition 2.5 | `operatorIneq_of_choiTwoBound`, `operatorIneq_of_FGP` |
| Remark 2.7, $r$-block positivity | `isBlockPositive_psiChoi` |
| Equation (16) | `Ppos_sub_Pneg` |
| Equation (19) | `Phi4_apply` |
| Equation (20) | `Phi4_ptransposeUV_rankOne` |
| Equation (21) | `HopScaled_eq` |
| Equation (26) | `Tsyn_eq_delta_vtx` |
| Equation (30) | `norm_Tsyn_sq_le` |
| Equation (31) | `inner_delta_placeQ_kron_zetaV` |
| Lemma 3.1 | `contraction_norm`, `contraction_trace`, `hsNormSq_ptraceU_rankFactor`, `hsNormSq_ptraceV_rankFactor` |
| Equation (40) | `exists_rankFactor` |
| Equation (43) | `rank_mono` |
| strict case of Equation (43) | `rank_mono_strict` |
| Equation (2) | `rank_r_partial_trace_exact` |
| strictness for rank $C < r$ | `rank_r_partial_trace_strict` |
| equality forces rank $C = r$ | `rank_eq_of_eq_rank_r_partial_trace` |
| trace–rank bound in Section 3 | `normSq_trace_le` |
| Lemma 4.7 | `hsNormSq_ptraceU_le`, `hsNormSq_ptraceV_le` |
| Equation (55) | `ptraceU_rankOne`, `ptraceV_rankOne` |
| extremizer $P_r \otimes \lvert v\rangle\langle w\rvert$ | `projWit` |
| its rank and its four norms | `rank_projWit`, `hsNormSq_projWit`, `trace_projWit`, `hsNormSq_ptraceU_projWit`, `hsNormSq_ptraceV_projWit` |
| equality at the extremizer | `projWit_bound_eq` |
| optimality of $a = r$ | `le_coeff_hsNormSq_of_bound` |
| optimality of $b = 1/r$ | `inv_le_coeff_trace_of_bound` |

Three respects in which the formal statements are renderings rather than transcriptions should be recorded. First, `FGPBound` — used only for the bridge theorems above — is stated in the homogeneous form Equation (10) rather than for unit vectors, and quantifies over expressions $|u_1\rangle\langle v_1| + |u_2\rangle\langle v_2|$ rather than over vectors of Schmidt rank at most two; by Equation (1) these are the same statement. Second, the formal analogues of $H_r$, $\Phi$ and $P_\pm$ carry positive integer scalings, chosen so that no denominators occur; this does not affect positivity. Third, what is certified is the deduction, not the faithfulness of the formal definitions to the informal ones; the conventions those definitions encode are exactly the ones fixed in the introduction.

The formalization follows the arguments of Sections 2 and 3 except at two points, where it takes a shorter route: Equation (24) is replaced by a Bessel-type duality that never forms $\mathcal{T}$ as a matrix, and the complete-graph identity of Proposition B.2 is not used, the vertex Cauchy–Schwarz estimate sufficing. Those two displays are therefore not themselves covered.

Of Section 4, only Lemma 4.7 is covered, in the elementary form proved there rather than through trace-norm duality; the Schatten-1 norm appears nowhere in the development. The remaining results of Section 4, and those of Appendix B and Appendix C, are not covered.

Theorem 2.4 *is* covered, in both operator forms and for an arbitrary transpose-symmetric Kraus family: `Choi.lean` carries Equation (28) with $(\Phi, \beta_2)$ abstract and `Lifting.lean` carries Equations (12) and (13), so $\Phi = \Lambda_U \otimes \Lambda_V$ enters only through Equation (11). Consequently Equation (44) is covered at every $\beta_2$, not only at $\beta_2 = 1$. Equation (33) is covered as `Psi_eq_tensor_square`, with the two structural inputs of Remark 2.7 — monoidality of $\Delta$ and Equation (32) — as separate lemmas, and the $r$-block-positivity reading of Remark 2.7 as `isBlockPositive_psiChoi`. The $r$-positivity statement of Theorem 2.4 is formalized in the unrolled form $\operatorname{rank} C \le r$ rather than through a definition of Schmidt rank, exactly as `FGPBound` is. Remark 2.6 and Remark 2.8 are observations about the mechanism of the formalized proof, not further claims; the $\mathbb{Z}/2$-grading and the two-dimensionality of the commutant are not themselves formalized.

## Appendix B. Gram structure and defect identities

This section records the algebraic defect structure of the rank-$r$ proof. The main residual is represented by positive Gram operators once the range isometry is fixed; this is not claimed to be a uniform bounded-degree polynomial sum-of-squares identity in the raw Stiefel coordinates. The only literal polynomial SOS below is the elementary trace–rank residual.

For an operator $C$ of rank at most $r$, define

$$\mathsf{G}_r(C) := r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \|\operatorname{Tr}_U C\|_2^2 - \|\operatorname{Tr}_V C\|_2^2, \tag{66}$$

$$\mathsf{R}_r(C) := r\|C\|_2^2 - |\operatorname{Tr} C|^2. \tag{67}$$

**Proposition B.1** (Polynomial SOS for the trace–rank residual)**.** *Assume $r \le \dim(U \otimes V)$ and write*

$$C = \sum_{i=1}^{r} |e_i\rangle\langle d_i|,$$

*where $e_1, \dots, e_r$ are orthonormal and zero columns $d_i = 0$ are allowed. In $(U \otimes V) \otimes \mathbb{C}^r$, put*

$$\mathbf{e} := \sum_i e_i \otimes q_i, \qquad \mathbf{d} := \sum_i d_i \otimes q_i.$$

*Equip the exterior square with the convention that $u_\alpha \wedge u_\beta$ for $\alpha < \beta$ is orthonormal whenever $(u_\alpha)$ is an orthonormal basis. Then*

$$\mathsf{R}_r(C) = \|\mathbf{e} \wedge \mathbf{d}\|^2. \tag{68}$$

*Equivalently, in any orthonormal coordinate system,*

$$\mathsf{R}_r(C) = \sum_{\alpha<\beta} |\mathbf{e}_\alpha \mathbf{d}_\beta - \mathbf{e}_\beta \mathbf{d}_\alpha|^2. \tag{69}$$

*Proof.* One has

$$\|\mathbf{e}\|^2 = r, \qquad \|\mathbf{d}\|^2 = \|C\|_2^2, \qquad |\langle \mathbf{e}, \mathbf{d}\rangle| = |\operatorname{Tr} C|.$$

The complex Lagrange identity gives both formulas. ☐

For the remainder of this section, fix an exact rank-$r$ factorization

$$C = \sum_{i=1}^{r} |e_i\rangle\langle d_i|, \qquad \delta_e = \sum_i e_i \otimes q_i, \qquad \delta_d = \sum_i d_i \otimes q_i,$$

and retain $\rho_0$, $\Pi_e$, $P_\pm$, and $\Phi$ from the proof of Proposition 2.5. Let $\mathcal{T}_- := \mathcal{T}$ be the negative-sector synthesis map in Equation (23). Choose an orthonormal basis of $\operatorname{ran} P_+$ and let $\mathcal{T}_+$ be the synthesis map whose columns are all Kraus images under $\Phi$ of those positive-sector basis vectors. Then

$$\Phi(P_\pm) = \mathcal{T}_\pm \mathcal{T}_\pm^*.$$

For a coefficient array $c = (c^{(ij)}_{\alpha\beta})$, define

$$K_{ij} := \sum_{\alpha,\beta} c^{(ij)}_{\alpha\beta} L_\alpha \otimes M_\beta.$$

At a vertex $k$, associate to an incident edge the signed vectors

$$v_{k,ik} := K_{ik}\bar{e}_i \quad (i<k), \qquad v_{k,kj} := -K_{kj}\bar{e}_j \quad (k<j).$$

**Proposition B.2** (Complete-graph defect certificate)**.** *With the skew Kraus normalization used in Equation (25),*

$$\boxed{\begin{aligned}&(r-1)\|c\|_2^2 - \|\mathcal{T}_- c\|^2\\ &\quad = \frac{r-1}{2}\sum_{i<j}\left[\frac{1}{2}\|K_{ij}\|_2^2 - \|K_{ij}\bar{e}_i\|^2 - \|K_{ij}\bar{e}_j\|^2\right]\\ &\qquad + \frac{1}{2}\sum_{k=1}^{r}\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e}-v_{k,f}\|^2.\end{aligned}} \tag{70}$$

*Consequently* $\|\mathcal{T}_-\|^2 \le r-1$ *and*

$$(r-1)(I-\Pi_e) - \mathcal{T}_-\mathcal{T}_-^* \succeq 0.$$

*Proof.* At every vertex use the exact variance identity

$$(r-1)\sum_{e\ni k}\|v_{k,e}\|^2 - \|\sum_{e\ni k} v_{k,e}\|^2 = \sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e}-v_{k,f}\|^2.$$

By Equations (25) and (26),

$$\begin{aligned}&(r-1)\|c\|_2^2 - \|\mathcal{T}_- c\|^2\\ &\quad = \frac{r-1}{4}\sum_{i<j}\|K_{ij}\|_2^2 - \frac{1}{2}\sum_{k=1}^{r}\|\sum_{e\ni k} v_{k,e}\|^2\\ &\quad = \frac{r-1}{4}\sum_{i<j}\|K_{ij}\|_2^2 - \frac{r-1}{2}\sum_{i<j}\left(\|K_{ij}\bar{e}_i\|^2 + \|K_{ij}\bar{e}_j\|^2\right)\\ &\qquad + \frac{1}{2}\sum_{k=1}^{r}\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e}-v_{k,f}\|^2.\end{aligned}$$

Factoring the first two sums gives exactly Equation (70). The vertex-variance terms are nonnegative, and each edge bracket is nonnegative by Lemma 2.1. Hence $\|\mathcal{T}_-\|^2 \le r-1$. Finally, Equation (31) gives $\operatorname{ran}\mathcal{T}_- \subseteq \delta_e^\perp$, whose projection is $I-\Pi_e$, so the operator inequality follows. ☐

**Proposition B.3** (Exact positive/negative-sector Gram identity)**.** *Put*

$$B_E := (r-1)(I-\Pi_e) - \mathcal{T}_-\mathcal{T}_-^*. \tag{71}$$

*Then*

$$\boxed{\mathsf{G}_r(C) = \|\mathcal{T}_+^*\delta_d\|^2 + \langle \delta_d, B_E\delta_d\rangle.} \tag{72}$$

*Moreover,* $B_E \succeq 0$.

*Proof.* By Lemma 3.1, $\mathsf{G}_r(C) = \langle \delta_d, H_r^{(0)}\delta_d\rangle$. Substituting Equation (21) and the two synthesis factorizations gives Equation (72). Positivity of $B_E$ is the operator conclusion of Proposition B.2. ☐

*Remark* B.4 (Scope of the Gram certificate)*.* The complete-graph identity separates the global synthesis-map defect into vertex variances and edgewise double-skew deficits. Edgewise nonnegativity is precisely the Fu–Gao–Park projection estimate used in Lemma 2.1. No uniform polynomial SOS certificate in the raw range-isometry variables is claimed.

## Appendix C. Generic consequences of $r$-block positivity

The two results collected here are separated from Section 4 because neither uses any property of the witnesses constructed there beyond $r$-block positivity, and Proposition C.3 does not use Theorem 1.1 at all. They are recorded because they are what makes those witnesses usable in practice.

### C.1. **Matrix-valued Schmidt-number criteria.**

**Corollary C.1** (A positive-map semidefinite cut)**.** *Let $1 \le r \le d^2$, let $X$ be a finite-dimensional Hilbert space, and put*

$$s := \min\{r,d\}, \qquad \Lambda_r := \mathcal{R}_{-1/s}^{\otimes 2}.$$

*If a positive semidefinite operator $\sigma \in \mathcal{L}(X \otimes (\mathbb{C}^d)^{\otimes 2})$ has Schmidt number at most $r$ across $X : (\mathbb{C}^d)^{\otimes 2}$, then*

$$(\mathrm{id}_X \otimes \Lambda_r)(\sigma) \succeq 0. \tag{73}$$

*More generally, Equation (73) holds with $\Lambda_r$ replaced by $\mathcal{R}_{-a}\otimes\mathcal{R}_{-b}$ whenever $a,b \ge 0$ and*

$$\max\{a,b\} \le \frac{1}{\min\{r,d\}}.$$

*Proof.* By Corollary 4.9, the indicated map is $r$-positive. Decompose $\sigma$ into positive multiples of pure states of Schmidt rank at most $r$. For each such vector, its Schmidt support on $X$ has dimension at most $r$, so positivity follows from $r$-positivity after restricting the identity ancilla to that support. Conjugating back into $X$ and summing the resulting positive operators proves the assertion. ☐

*Remark* C.2 (Use as an SDP cut)*.* Whenever a semidefinite relaxation contains an explicit candidate state block $\sigma$, Equation (73) is a linear matrix inequality in the entries of that block. If those entries are affine functions of a moment matrix, applying $\mathrm{id} \otimes \Lambda_r$ produces an affine matrix pencil and therefore a valid additional semidefinite constraint under a Schmidt-number-$r$ promise. This is a necessary condition for that promise, not in general a complete semidefinite characterization of Schmidt number at most $r$.

C.2. **Local aggregation of rank witnesses.**

**Proposition C.3** (Local-channel aggregation). *Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be finite-dimensional Hilbert spaces. For each $e$ in a finite set $E$, let*

$$\Phi_e^A : \mathcal{L}(\mathcal{H}_A) \to \mathcal{L}(\mathcal{K}_{A,e}), \qquad \Phi_e^B : \mathcal{L}(\mathcal{H}_B) \to \mathcal{L}(\mathcal{K}_{B,e})$$

*be quantum channels, let $W_e = W_e^* \in \mathcal{L}(\mathcal{K}_{A,e} \otimes \mathcal{K}_{B,e})$ be $r$-block-positive, and let $\lambda_e \geq 0$. Then*

$$H := \sum_{e \in E} \lambda_e (\Phi_e^A \otimes \Phi_e^B)^*(W_e) \tag{74}$$

*is $r$-block-positive across $\mathcal{H}_A : \mathcal{H}_B$. In particular,*

$$\mathrm{Tr}(H\rho) \geq 0 \qquad \textit{whenever } \mathrm{SN}(\rho) \leq r.$$

*Proof.* Local quantum channels do not increase Schmidt number. Indeed, write

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \qquad \mathrm{SR}(\psi_k) \leq r,$$

and choose Kraus operators $A_{e,i}$ and $B_{e,j}$ for $\Phi_e^A$ and $\Phi_e^B$. Then

$$(\Phi_e^A \otimes \Phi_e^B)(\rho) = \sum_{k,i,j} p_k |(A_{e,i} \otimes B_{e,j})\psi_k\rangle\langle(A_{e,i} \otimes B_{e,j})\psi_k|,$$

and every vector in this decomposition has Schmidt rank at most $r$. Hence

$$\mathrm{SN}\big((\Phi_e^A \otimes \Phi_e^B)(\rho)\big) \leq r$$

for every $e$. Therefore

$$\begin{aligned} \mathrm{Tr}(H\rho) &= \sum_{e \in E} \lambda_e \, \mathrm{Tr}\left[W_e (\Phi_e^A \otimes \Phi_e^B)(\rho)\right] \\ &\geq 0. \end{aligned}$$

□

**Corollary C.4** (A local-Hamiltonian Schmidt-number certificate). *In Proposition C.3, suppose each local output contains two matched $d_e$-dimensional pairs, with $r \leq d_e$, and take*

$$W_e = \left(I - \frac{1}{r}|\Omega_e\rangle\langle\Omega_e|\right)^{\otimes 2}.$$

*Then the operator $H$ in Equation (74) satisfies*

$$\mathrm{Tr}(H\rho) < 0 \quad \Longrightarrow \quad \mathrm{SN}(\rho) \geq r + 1.$$

*For literal subsystem tests, the channels $\Phi_e^A$ and $\Phi_e^B$ may be taken to be the corresponding partial traces, so $H$ is a sum of local terms embedded into the global Hilbert space. More generally, any geometric locality of $H$ is inherited from the subsystem supports of the chosen channels; arbitrary channels need not produce local Hamiltonian terms.*

*Remark* C.5 (Positive-semidefinite pulled-back terms). Write

$$\Psi_e := \Phi_e^A \otimes \Phi_e^B$$

and shift the output-space witness to

$$h_e^{\mathrm{out}} := W_e - \lambda_{\min}(W_e) I_e \succeq 0,$$

where $I_e$ is the identity on $\mathcal{K}_{A,e} \otimes \mathcal{K}_{B,e}$. Its global pullback

$$\widetilde{h}_e := \Psi_e^*(h_e^{\mathrm{out}}) \succeq 0$$

is well defined even when the output spaces depend on $e$. Since $\Psi_e$ is trace-preserving, $\Psi_e^*(I_e) = I$, and hence

$$\sum_e \lambda_e \widetilde{h}_e = H - \left( \sum_e \lambda_e \lambda_{\min}(W_e) \right) I.$$

Consequently, every normalized $\rho$ with $\mathrm{SN}(\rho) \le r$ satisfies

$$\mathrm{Tr}\left[ \sum_e \lambda_e \widetilde{h}_e \, \rho \right] \ge - \sum_e \lambda_e \lambda_{\min}(W_e).$$

For the witnesses in Corollary C.4,

$$\lambda_{\min}(W_e) = 1 - \frac{d_e}{r},$$

so the known energy baseline is

$$\sum_e \lambda_e \left( \frac{d_e}{r} - 1 \right).$$

This construction certifies Schmidt number across the prescribed bipartition; by itself it is not an NLTS statement.

## References


[1] Man-Duen Choi. "Completely positive linear maps on complex matrices." Linear Algebra and its Applications, 10:285–290, 1975. https://doi.org/10.1016/0024-3795(75)90075-0

[2] Pablo Costa Rico. "New partial trace inequalities and distillability of Werner states." Letters in Mathematical Physics, 115:47, 2025. https://doi.org/10.1007/s11005-025-01935-y

[3] Pablo Costa Rico and Michael M. Wolf. "Partial trace relations beyond normal matrices." arXiv:2507.18278v1, July 24, 2025. https://arxiv.org/abs/2507.18278v1

[4] Jinshi Fu, Li Gao, and Sang-Jun Park. "A solution to 2-copy distillability of Werner states." arXiv:2607.21367v1, July 23, 2026. https://arxiv.org/abs/2607.21367v1

[5] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*, 2nd edition. Cambridge University Press, 2012.

[6] Michał Horodecki, Paweł Horodecki, and Ryszard Horodecki. "Mixed-state entanglement and distillation: Is there a 'bound' entanglement in nature?" Physical Review Letters, 80:5239–5242, 1998. https://doi.org/10.1103/PhysRevLett.80.5239

[7] Nathaniel Johnston and David W. Kribs. "A family of norms with applications in quantum information theory." Journal of Mathematical Physics, 51:082202, 2010. https://doi.org/10.1063/1.3459068

[8] Barbara M. Terhal and Paweł Horodecki. "Schmidt number for density matrices." Physical Review A, 61:040301, 2000. https://doi.org/10.1103/PhysRevA.61.040301

[9] Reinhard F. Werner. "Quantum states with Einstein–Podolsky–Rosen correlations admitting a hidden-variable model." Physical Review A, 40:4277–4281, 1989. https://doi.org/10.1103/PhysRevA.40.4277